



**Verified Carbon
Standard**

TRAMONTANA ICS-01 – THE DISTRIBUTION OF IMPROVED COOKSTOVES TO INDIGENOUS COMMUNITIES ACROSS RURAL INDIA



Document Prepared by Earthood Services Private Limited

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Summary:

The Project Activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” involves distribution of fuel efficient improved cookstoves (ICS) directly to individual households in rural areas across multiple states in India. These ICS are distributed free of cost and the project proponent is using carbon financing as the means to overcome a key hurdle for their adoption. The Tramontana ICS to be distributed forms a safe, portable cleaner cooking technology which can be (voluntarily) adopted by the communities, with an increased thermal efficiency, and completeness of combustion through its energy-efficient design, technology and the materials used. GHG emissions are immediately reduced, and families experience significant improvements in the levels of toxic indoor air pollution. The improved efficiency lowers deforestation/forest degradation (and consequently biodiversity loss) and enhances the daily lives particularly of female family members, who are most involved in carrying out the daily tasks relating to the stove.

Earthood Services Private Limited (Earthood) has performed a joint validation and verification of the project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” by Tramontana Impact Limited. This assignment was performed based on the rules and requirements defined by VCS for the project activities. The review of this joint Project Description & Monitoring Report (PDMR), supporting documentation and subsequent follow-up actions (onsite visit and interviews with end-users and local stakeholders) have provided Earthood with sufficient evidence to determine the fulfilment of stated criteria. The project correctly applies the methodology VMR0006: Energy efficiency and fuel switch measures in thermal applications (Version 1.2)/07/ along with CDM AMS-II G: Energy efficiency measures in thermal applications of non-renewable biomass (Version 13.1)/08/

The project results in reductions of CO₂e emissions that are real, measurable and give long-term benefits to the mitigation of climate change. It is demonstrated that the project is not a likely baseline

scenario and the emission reductions attributable to the project are, hence, additional to any that would occur in the absence of the proposed VCS project activity. The average annual and total GHG emissions reductions from the project is expected to be 1,310,160 and 9,171,123 tCO₂e , respectively, over the first 7-year crediting period. The monitoring period identified for the first verification period is from 21/01/2023 to 23/08/2023, and the achieved emission reductions during this monitoring period is 62,743 tCO₂e.

The scope of the services provided by the Earthood Services Private Limited is to perform joint validation & verification of the project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India”. The scope is to assess the claims and assumptions made in the joint Project Description & Monitoring Report /01/ against the VCS criteria, UNFCCC criteria, including but not limited to the VCS Standard /03/, VCS Validation and Verification Manual /04/ and other relevant rules and requirements established by Verra.

This assignment is an independent and objective review for determination of the monitored reductions in GHG emissions. The validation & verification process is undertaken by the VVB which involves the following:

- The desk review of documents and evidence submitted by the project participant in the context of the VCS criteria.
- Undertaking onsite audit, interviews, or interactions with representatives of the project participant,
- Reporting audit findings with respect to clarification and non-conformities and the closure of the findings as appropriate.
- Preparing a draft joint validation & verification report under the VCS program complying with the VCS requirements
- Technical review of the draft validation & verification opinion along with other documents as appropriate by an independent competent technical review team for finalization of the joint validation & verification opinion (this report)
- Following acceptance of the final report by the Technical Reviewer it is then approved by Earthood Services Private Limited in accordance with the VCS and CDM procedures.

During joint validation and verification of the project, a total of 18 CLs, 1 CARs & 2 FARs were raised. These have been successfully resolved as provided in Appendix 5 of this report. VVB has undertaken an elaborate audit trail for validation of assumptions and claims contained in the final PDMR Version 1.5 dated 16/07/2024. Each detail has been checked during the site visit and document review which is not only confined to information provided by the Project Proponent but also through undertaking independent evaluations, including relevant publicly available information through utilizing the technical expertise of the assessment team. Thus, there are no evident uncertainties associated with the validation.

Summary of the validation and verification conclusions:

The review of the project description & monitoring report, supporting documentation and subsequent follow-up actions have provided Earthood with strong evidence to determine the fulfilment of all criteria. Earthood is of the opinion that the project activity “Tramontana ICS-01 – The Distribution of

Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID: 4457) strictly meets all of the VCS requirements and has correctly applied the CDM and VCS approved methodologies AMS-II. G: Energy efficiency measures in thermal applications of non- renewable biomass (Version 13.1) and VMR0006: Energy efficiency and fuel switch measures in thermal applications (Version 1.2) respectively. The GHG emission reductions were calculated correctly based on the approved methodologies. Earthood Services Private Limited can certify that the emission reductions achieved in the project description & monitoring period 21/01/2023 - 23/08/2023 by VCS project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” amount to 62,743 tCO₂e.

Therefore, this is being submitted for request for issuance, as per VCS and UNFCCC procedures. In summary, it is Earthood’s opinion that the project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” as described in the PDMR Version 1.5 dated 17/06/2024 meets all the relevant baseline and monitoring methodologies. The monitoring plan adequately provides for the ex-post monitoring of the project’s emission reductions. The monitoring arrangements described in the monitoring plan are feasible within the project design and it is Earthood’s opinion that the Tramontana Impact Limited shall be able to implement the monitoring plan and requirements of VCS standard. Therefore, Earthood requests the registration of the project as a VCS project activity.

CONTENTS

1	INTRODUCTION.....	6
1.1	Objective	6
1.2	Scope and Criteria.....	6
1.3	Reasonableness of Assumptions and Level of Assurance	7
1.4	Summary Description of the Project.....	8
2	VALIDATION AND VERIFICATION PROCESS.....	9
2.1	Method and Criteria	9
2.2	Document Review	10
2.3	Interviews	11
2.4	Site Visits	11
2.5	Resolution of Findings	14
3	VALIDATION FINDINGS.....	15
3.1	Project Details	16
3.2	Project Activity Instances in Grouped Projects	23
3.3	Safeguards.....	23
3.4	Application of Methodology	32
3.5	Non-Permanence Risk Analysis.....	46
4	VERIFICATION FINDINGS.....	46
4.1	Project Implementation Status.....	46
4.2	Accuracy of Reduction and Removal Calculations	46
4.3	Quality of Evidence to Determine Reductions and Removals	48
5	VALIDATION AND VERIFICATION OPINION	49
5.1	Validation and Verification Summary.....	49
5.2	Validation Conclusion.....	50
5.3	Verification conclusion	51
5.4	Ex-ante vs Ex-post ER Comparison.....	52
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION	54
	APPENDIX 2: LIST OF DOCUMENTS REFERRED	55
	APPENDIX 3: ABBREVIATIONS	58
	APPENDIX 4: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS.....	60
	APPENDIX 5: LIST OF FINDINGS	64

1 INTRODUCTION

1.1 Objective

Earthood Services Private Limited (VVB) has been contracted for joint validation & verification services for the VCS project “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (project ID VCS 4457) undertaken by Tramontana Impact Limited (PP), which is initially located in the state of Andhra Pradesh in India to jointly carry out the validation and verification of the project against the requirement of VCS Program.

The objective of the joint validation-verification assessment is to determine the

- Project conformance to VCS rules.
- Project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and
- Likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented.

To fulfil the above objectives, the validation/verification body (VVB) assigned the task of assessment to the validation team that is collectively qualified as per Earthood’s Quality Management System (QMS) manual and other internal requirements.

The validation/verification is an independent assessment of the project by a VVB that determines whether the project complies with the VCS rules and meets a satisfactory standard.

1.2 Scope and Criteria

The joint validation and verification scope of this assignment is defined as an independent and objective review of the VCS Project Description & Monitoring Report (VCS PDMR)/01/ and supporting annexures, which is reviewed by the assessment team against the relevant criteria and decisions by the VCS Board, including but not limited to, the approved baseline and monitoring methodology, applicable tools, standards (CDM and VCS) and guidance. The assessment team has, based on the recommendations prescribed in the latest version of VCS Validation and Verification Manual employed a risk-based approach, focusing on the identification of significant risks for project implementation and the generation of VCU.

The scope of this assignment is to:

- Jointly conduct validation and verification in accordance with Earthood’s own Quality management system that is based on the VCS VVM along with the guidance provided by the VCS board to determine if the project meets all applicable VCS requirements,

including those specified in the VCS standard, relevant methodologies, tools, and guidelines and processing the same with the VCS Program Guide.

- Assess the accuracy, conservativeness, relevance, completeness, consistency, and transparency of the information provided by the project proponent.
- Determine whether information provided by the project proponent is reliable and credible.
- Present information in the form of a validation & verification report in a factual, neutral, coherent manner and referencing the sources and assumptions, plus other forms of validation means employed.
- Report the findings and conclusions in an objective manner and conduct all validation & verification in accordance with VCS rules and procedures.
- Apply consistent validation and verification criteria in providing expert judgments to the requirements of applicable approved methodologies, tools, and also cross check the same.
- Safeguard the confidentiality of all information obtained during validation and verification; and
- Adhere to the principles of independence, ethical conduct, fair presentation, and due professional care in the assessment process.

1.3 Reasonableness of Assumptions and Level of Assurance

☐ Limited level of assurance

☒ Reasonable level of assurance

A draft joint validation and verification report that is prepared by the assessment team is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures established and implemented by Earthood are duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable VCS requirements as appropriate. The technical review team is collectively required to possess the technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of the technical review team are independent of the validation and verification team. The report approved by the Technical Manager is then endorsed by the Managing Director, who is overall responsible to ensure quality, before final release. The further details of applicable procedures and responsibilities about Earthood's Quality Management System (QMS) are available on its website (www.earthood.in).

Earthood's validation approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. The validation and

verification are planned and performed by obtaining evidence, additional information and explanations that the assessment team considers necessary to give reasonable assurance that the reported estimated GHG emission reductions are fairly stated and are in compliance with the standards and guidelines of Verra. A reasonable level of confidence is achieved after conducting a site visit, utilizing reputable and dependable methods for joint validation and verification, as elaborated in detail in section 2.4 below. All documentary evidence was checked, and an onsite audit was conducted accompanied by interviews of PP representatives, site personnel, other involved parties and local stakeholders to arrive at a validation conclusion by the assessment team.

In our opinion the estimated and actual GHG emissions reductions were calculated correctly based on the approved methodologies VMR0006: Energy efficiency and fuel switch measures in thermal applications (v1.2)/07/ and CDM AMS-II G: Energy efficiency measures in thermal applications of non-renewable biomass (v13.1)/08/. The assessment result has a reasonable level of assurance in validation & verification that GHG assertions are free of material errors, omissions and misrepresentations. The documents reviewed are included under Appendix 2 of this report.

1.4 Summary Description of the Project

A summary description of the project activity has been illustrated in section 1.1 of Joint PDMR/01/ appropriately. As per the given details in PD, information was assessed onsite and found to be consistent. The project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” involving distribution of fuel efficient improved cookstoves (ICS) covering multiple instances under a grouped project has been developed. The initial instance is located in the state of Andhra Pradesh, from which the project can expand to other states including Odisha, Telangana and Chhattisgarh. The first project instance will distribute approximately 50,000 ICS after which multiple instances are planned. The project is a voluntary initiative as host country laws do not mandate the distribution of ICS to the local communities. The project proponent is Tramontana Impact Limited who is using carbon finance to implement the project, involving different activities including production, distribution, and maintenance of various product technologies. This project includes technologies, designed to reduce GHG emissions by supplying high efficiency ICS which enable cleaner cooking. The scenario is well described and consistent with the data that was verified during the onsite audit. The average annual GHG emission reduction from the project is expected to be 1,310,160 tCO₂e over the first 7-year crediting period from 21/01/2023 to 20/01/2030. The monitoring period identified for the first verification period is from 21/01/2023 to 23/08/2023, and the achieved emission reductions during this monitoring period is 62,743 tCO₂e.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The validation of the crediting period process is conducted as per internal Earthood Quality Manual and in accordance with criteria laid down by ISO 14064-2 & VCS. It includes the following steps:

- contract with PP for the scope and appointment of validation team and technical review team.
- completeness check of VCS-PDMR/01/
- desk review of VCS-PDMR/01/ and estimated GHG emission reduction calculation sheet/13/
- onsite inspection and interviews by validation team/19/
- follow up interviews with PP and Project Owner
- reporting and closure of findings (CARs/CLs/FARs) and preparation of draft validation report
- independent technical review: The project documentation is reviewed by internal technical reviewer in order to independently confirm whether the applicable GHG program requirements were objectively met or not, in addition to whether internal procedures were followed while arriving at the validation and verification opinion. The technical reviewer may accept or reject the validation and verification opinion prepared by the assessment team and give the reasons. The resolved findings may be opened at this stage or new findings may be identified that are required to be addressed by assessment team and/or project proponents, as appropriate.
- issuance of the final validation/verification report to the PP

Earthood keeps all documents and records in a secure and retrievable manner for at least two years after the end of the project crediting period.

Sampling plan for validation of the project:

The VVB has followed Standard for Sampling and surveys for CDM project activities and programmes of activities, Version 9/12/, and applied acceptance sampling in the validation in accordance with para 28. Para 30 and 31 of the standard suggests the AQL and UQL and maximum errors which have been followed by the VVB. Table 2 of Standard for Sampling and

surveys for CDM project activities and programmes of activities, Version 9, prescribes minimum sample size and acceptance number for discrepant values.

Applying para 30 and 31 of the standard, validation will require 11 samples (end users) to validate the project developer's sampling plan.

Reference number	AQL	UQL	Producer Risk	Consumer Risk	Sample Size; Min	Acceptance No.
VCS 4457	0.5%	20%	10%	10%	11	0

Sampling plan for verification of the project:

A similar approach for the verification was applied by the VVB in line with CDM Sampling Standard/12/. 11 samples were picked by the VVB for interview.

VVB has interviewed a total 22 samples (11 each for validation and verification respectively) to verify the consistency, accuracy, and completeness, of the implementation of the project activity as cited in the PDMR/1/. The assessment team who visited the site collectively had the competence for the relevant technical area to which the project relates. The validation and verification opinion is therefore not only informed by a desk review, but also by views collected from the end users of the technology during an extensive on-site visit.

2.2 Document Review

This joint validation & verification of the project is performed primarily as a document review of the VCS Joint PDMR /01/ and associated documents as stated in detail in appendix 2 of this document. The assessment is performed by a validation & verification team using dedicated protocols. The assessment team cross checks the information provided in documents including the VCS Joint PDMR, and available information from sources other than those used based on the team's sectoral or local expertise, and also conducts independent background investigations. Earthood conducted a desk review as follows:

- a) A review of the data and information presented to verify their completeness.
- b) A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures

- c) An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

A list of documents reviewed during this verification are annexed as Appendix 2 in this report.

2.3 Interviews

The VVB conducted an onsite visit from 15/10/2023 to 17/10/2023. In accordance with paras 4.1.11 and 4.1.12 of the VCS Standard v4.7/03/, a site visit to facilities and/or project areas shall be conducted at validation, project crediting period renewal validation, the first verification of the project after validation, verifications that include project baseline reassessments; and verifications that assess a project description deviation where the deviation impacts the applicability of the methodology, additionality or the appropriateness of the baseline scenario. As the project activity is at validation and first verification stage, an onsite visit has been conducted by the VVB.

During the onsite visit the following points were discussed as part of the interviews of project representatives:

- a) The introduction, scope, and objective of the work, along with the roles and responsibilities of the audit team, necessary resources, and the onsite audit timetable, were addressed. Any concerns expressed by the Project Proponent (PP) were considered and incorporated into the planning process.
- b) A comprehensive validation of all details pertaining to the project activity, as outlined in the Joint PDMR/01/, was conducted through in-depth discussions with the PP. This approach ensured high quality in cross-referencing project information.
- c) Usage of project device and baseline device as part of end-user interviews. A total 22 random samples were chosen as per the sampling plan explained in section 2.1, out of which 11 were baseline samples and 11 were monitoring samples. The assessment team interviewed the end-users /19/.
- d) A thorough evaluation was conducted to verify the compliance of monitoring procedures at the project site with the registered VCS PD and monitoring methodology. This review aimed at assessing accuracy and traceability in the monitoring process, including the impact of the project on Sustainable Development Goals (SDGs).

Assessment team also interviewed local stakeholders who were part of the LSC meetings/09/. And all the details of interviews conducted during the onsite visit have been provided in section 2.4.

2.4 Site Visits

The VVB carried out an onsite visit from 15/10/2023 to 17/10/2023 and physically inspected the project technical design and implementation as specified in the VCS PDMR /01/.

The onsite visit was conducted to validate and verify the accuracy and completeness of the project description. The assessment team members involved in the onsite audit collectively had the competence for the relevant technical area to which the project relates. The observations from the onsite audit have been considered while concluding the validation and verification opinion.

As discussed in section 2.3, the details of multiple interviews conducted during the onsite inspection have been given in detail below:

Name	Affiliation	Date (Opening Meeting)	Date (Closing Meeting)	Assessment team
Ramanan Loganathan	Tramontana	15/10/2023	17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Saibaba Abburi	Cropzone	15/10/2023	17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Narasimha Barpata	Cropzone	15/10/2023	17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Ramamurthy Dommaraju	Cropzone	15/10/2023	17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala

Local stakeholders interviewed during the site visit:

Name	Affiliation	Date	Assessment team
Kuda Mani	Local Stakeholder	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Kuda Sanjana	Local Stakeholder	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Vantala Karramma	Local Stakeholder	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Bandi Laxmi	Local Stakeholder	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala
Gadugu Chandramma	Local Stakeholder	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria, Nikhitha Chinthala

Baseline end-users interviewed during the Site visit:

Name	Affiliation	Date	Assessment team
Chompi Dharma Naidu	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kancham Bojjayya	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Seedhari Chiranjeevi	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Mottadam Rathna Kumari	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kurada Sravani	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Vanthala Narasinga Rao	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Muvvala Vijaya	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Ulli Meri	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kuda Narayanamma	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Pothuru Varalaxmi	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kanthamma	End-User-Baseline	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala

Monitoring end users interviewed during the site visit:

Name	Affiliation	Date	Assessment team
Pangi Ramanna	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Vanugu Padma Kumari	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Mori Chinnami	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala

Pangi Lakshmi	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kakarla Arjun	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Janni Baghya	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Killo Janaki	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Kavarla Devaki	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Killo Radhamma	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Killo Chinnalamma	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala
Korra Ramakrishna	End-User-Monitoring	15/10/2023 – 17/10/2023	Sushant Vashisht, Shifali Guleria & Nikhitha Chinthala

2.5 Resolution of Findings

The validation and verification protocols used for the assessment are designed in accordance with the latest version of VCS Program Documents and relevant CDM Standards. The objective of this step is to identify, discuss and conclude on the issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on a desk review and onsite assessment. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and nonconformities.

As an outcome of the joint validation and verification process, the assessment team can raise different types of findings. Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS/CDM requirements have been met. Where a non-conformance arises the validator/verifier raised a Corrective Action Request (CAR).

More specifically, a CAR is issued where either:

- The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions.

- The CDM/VCS requirements have not been met.
- There is a risk that emission reductions cannot be monitored or calculated.

A Forward Action Request (FAR) is raised during validation/verification to highlight issues related to project implementation that require review during the next verification of the project activity.

Irrespective of the nature of the findings, all of these are given to PP in a separate findings document. In the document, the project proponent(s) is/are given the opportunity to respond to the findings. Based on the responses by the PP along with relevant supporting documents/evidence, the validation team determines whether the findings are resolved or not. The findings may also be raised at later stages e.g., during the internal technical review process and these are also communicated/dealt in the same manner as described above. The draft joint validation & verification report is also prepared during this period reflecting the status of the findings, if any, as appropriate.

A total of 1 CAR, 18 CLs & 2 FARs were raised in the current joint validation and verification. All the findings that are raised and communicated to project participant during the verification are included under Appendix 4. The section also includes the responses, if provided, by the project participants and an assessment by the VVB team if it was closed out or otherwise. All CARs and CLs raised by the Earthood during verification shall be resolved prior to submitting a request for issuance.

In summary, the type and total number of findings that were raised are indicated below.

Type of Finding	CL	CAR	FAR
Total Findings	18	1	2

2.5.1 Forward Action Requests

Two Forward Action Requests (FARs) were raised during this assessment.

1. Ex-ante Emission Reductions provided on the Verra page and in the PDMR are inconsistent. The estimated ERs are to be made consistent between the project documents and project activity web page. See Table 3 for further details. VVB to verify the same in the subsequent verification.
2. For the current project activity instance the Alluri Seetharama Raju district has been covered in the baseline survey. A baseline survey has to be conducted before the implementation of any new PAIs in compliance with the applied methodology section 9.1.

3 VALIDATION FINDINGS

3.1 Project Details

The project “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” is appropriately selected in the VCS PDMR/01/, and the project category and type are correctly indicated in the VCS PDMR/01/. Therefore, VVB confirms that the information provided in the PDMR is complete and accurate.

Item	Evidence gathering activities, evidence checked, and assessment conclusion					
Audit history	Audit type	Period	Program	Validation/ verification body name	Number of years	
	Joint Validation + Verification	21-January-2023 – 23-August-2023	VCS	Earthood Services Private Limited	0.586	
Sectoral scope	03- Energy Demand					
AFOLU project category, if applicable	N/A					
Project activity type	Project Type: II – Energy Efficiency Improvement Project Grouped Project: Yes					
General eligibility of the project to participate in the VCS Program	The project activity involves energy-efficient cookstove distribution in rural areas of India which is not excluded as per the section 2.1 of the VCS standard v4.7/03/. The project is listed on the project pipeline which has been ensured by the VVB during the opening meeting with the PP before beginning the validation in line with para 4.1.5 of the VCS standard v4.7/03/. The methodology VMR0006 v1.2 /07/ is applicable for the project activity.					
AFOLU project eligibility, if applicable	N/A					
Transfer project eligibility, if applicable	N/A					

Project design

The project is a grouped project. For the inclusion of new project activity instances, the PP must ensure the following eligibility criteria from the VCS standard v4.7/03/.

Eligibility Criteria	Compliance by new project instances
Meet the applicability conditions set out in the methodology applied to the project	All new instances under the grouped project will meet the conditions of the methodology as specified in section 3.2
Use the technologies or measures specified in the project description	The technology type employed is an energy efficient ICS to replace the traditional inefficient cookstoves used by households. Information on the ICS technology used for all instances is provided in the form of manufacturer product specifications/18/, independent third-party certification/15/ for ICS thermal efficiency and through distribution data/photos/24/
Apply the technologies or measures in the same manner as specified in the project description	Only thermally efficient ICS will be distributed to households under project instances, and no other technologies will be adopted
Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area	All new project instances will remain within the geographical boundaries of India and will therefore share some similar characteristics. However, given in each instance corresponds to a different region, separate baseline campaigns are conducted to ensure a more accurate representation of the respective region. The baseline

		scenario, including for the first instance, is discussed in section 3.4. Baseline data for the first instance in ASR, Andhra Pradesh is provided by the PP and assessed by the VVB. For future PAls the Baseline survey will be conducted before the implementation of new PAls and reviewed under subsequent verifications, in compliance with the applied methodology section 9.1. FAR#2 has been raised.
	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area	All new project instances will have additionality characteristics consistent with the initial instance, and this will be demonstrated using the activity methodology below. Step 1: Regulatory Surplus Any new project instances included will not be mandated by any law, government policy or regulation in the host country. Step 2: Positive List The inclusion of new project instances will comply with the positive list under the methodology (see section 3.5 for details) Step 3: Project Method Not applicable
	The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity	VVB confirms that the PP has included wording in section 1.4 of the JPD-MR that each added project activity instance of the same project activity may not be spread across more than one project if they are within ten kilometres of each other as per section 3.6.8 of the VCS Standard v.4.7.

	and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other)	The VVB has validated the requirement based on the geolocations of beneficiaries recorded in the project database, and confirm that they are all within a single project
Project ownership	Tramontana Impact Limited (subsidiary of Tramontana Asset Management) is the Project Proponent. The project ownership is with the PP. All rights to carbon credits (VCUs) generated from project instances will be with the PP by means of a standardised license agreement physically signed by the beneficiary of each ICS. The signed agreement between the beneficiary and the PP has been verified from the distribution database/22/.	
Project start date	21-January-2023. The start date represents the distribution of the first ICS, and this has been verified from the distribution database/24/ provided by the PP which contains a substantial amount of objective evidence including date and time of submission, ICS serial numbers, signed user agreements, National ID proof of end user, Photographs of End User, Below poverty line card issued by Government of Andhra Pradesh under National food security Act, 2013, Photographs with their new ICS, and personal identification records. VVB also checked the Android app which was built by the PP to automatically record data including the date and time of entry which is automatically synced to the cloud from the local tablet. To further verify the project start date, the metadata attached to the photos (beneficiary with ICS, national ID card front and back, BPL card front and back, community certificate front and back) from the first recorded entry which was synced from the tablets was assessed. The VVB is providing this metadata to Verra in an Excel spreadsheet. The data confirms that the start date and time of creation of these photos was the 21st January 2023 (column E in the metadata spreadsheet), therefore proving the project activity's start date of 21st January 2023.	

Project crediting period	21-January-2023 to 20-January-2030, Project crediting period is seven years and renewable twice as per the VCS standard v4.7/03/.																				
Project scale	Large scale project, as the estimated emissions reductions are 1,310,160 tCO2e/year which is more than 300,000 tCO2e/year threshold. The project activity comes under large project as per the VCS Standard v4.7/03/.																				
Likelihood of achieving estimated GHG emission reduction or removals	High probability that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented.																				
Technologies and measures implemented by the project activity	Tramontana ICSs are being distributed in the project activity with the specifications given below which are verified from the manufacturer brochure/18/. These specifications will be used to determine emission reductions for the project. <table> <tr> <td>ICS Model</td><td>CRS-23</td></tr> <tr> <td>Type</td><td>Natural Draft</td></tr> <tr> <td>Thermal Efficiency</td><td>36.01%</td></tr> <tr> <td>Height</td><td>25.5 cm</td></tr> <tr> <td>Diameter</td><td>27 cm</td></tr> <tr> <td>Weight</td><td>9.471 kg</td></tr> <tr> <td>Eligible Fuel Type</td><td>Wood, Charcoal, Other Biomass</td></tr> <tr> <td>Portable or Fixed</td><td>Portable</td></tr> <tr> <td>Colour / Finish</td><td>Black & White / Glossy</td></tr> <tr> <td>Expected Lifespan</td><td>7 years</td></tr> </table> CL 06 has been raised and resolved.	ICS Model	CRS-23	Type	Natural Draft	Thermal Efficiency	36.01%	Height	25.5 cm	Diameter	27 cm	Weight	9.471 kg	Eligible Fuel Type	Wood, Charcoal, Other Biomass	Portable or Fixed	Portable	Colour / Finish	Black & White / Glossy	Expected Lifespan	7 years
ICS Model	CRS-23																				
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Eligible Fuel Type	Wood, Charcoal, Other Biomass																				
Portable or Fixed	Portable																				
Colour / Finish	Black & White / Glossy																				
Expected Lifespan	7 years																				

Implementation schedule of the project activity or activities	Tramontana has identified multiple target areas across several states of India which suffer from local deforestation and are primarily populated with indigenous communities. A high proportion of the families in the project area own a Below Poverty Line (BPL) or ration support card issued by the government which indicates the maximum annual income of Rs 60,000. This has been verified from the ration card and Aadhar card photos provided by the PP in the distribution database/24/. A distribution plan covering multiple instances under a group project has been developed by the PP. The initial instance is located in the state of Andhra Pradesh, from which the project can expand to other states including Odisha, Telangana and Chhattisgarh. <table data-bbox="850 747 1230 1224"> <tr> <th>Instance</th><th>Planned ICS Distribution</th></tr> <tr> <td>Region 1</td><td>50,000</td></tr> <tr> <td>Region 2</td><td>100,000</td></tr> <tr> <td>Region 3</td><td>100,000</td></tr> <tr> <td>Region 4</td><td>100,000</td></tr> <tr> <td>Region 5</td><td>50,000</td></tr> </table> As a part of first instance 46,160 ICSs were installed during the first monitoring period i.e. from 21-January-2023 to 23-August-2023.	Instance	Planned ICS Distribution	Region 1	50,000	Region 2	100,000	Region 3	100,000	Region 4	100,000	Region 5	50,000
Instance	Planned ICS Distribution												
Region 1	50,000												
Region 2	100,000												
Region 3	100,000												
Region 4	100,000												
Region 5	50,000												
Project location	The first project instance is within the Alluri Sitarama Raju (ASR) district of the state of Andhra Pradesh, with approximate coordinates of latitude 18.011959 and longitude 82.555030 and this has been verified from the Onsite visit/19/.												
Conditions prior to project initiation	The pre-project baseline scenario is the continued use of rudimentary traditional cookstoves by the communities, typically in the form of fixed mud clay stoves or three-stone stoves which has been confirmed by VVB from the interviews with the end users during the onsite visit/19/. It is also confirmed that the usage of these stoves has been commonplace in the project areas for generations, where												

	households burn woody biomass to meet their thermal energy needs including for the purposes of cooking and boiling water.
Project compliance with applicable laws, statutes and other regulatory frameworks	There exists no mandatory law or regulations requiring the introduction or use of ICS in India. The project is undertaken on a voluntary basis by the Project Proponent.
Double counting and participation under other GHG programs	PP has declared in the joint PDMR/01/ that the project has neither received nor seeking credit for reductions and removals from the project activity under another GHG program. VVB has checked the other GHG programs and verified that the PA has not been registered under any other GHG program. PP has also provided the declaration on no double counting/05/.
No double claiming with emissions trading programs or binding emission limits	Project emissions reductions and removals or project activities are not included in any emissions trading program or binding emission limit.
No double claiming with other forms of environmental credit	Project activity has not sought, received, nor is planning to receive credit from another GHG-related environmental credit system.
Supply chain (Scope 3) emissions double claiming	The project activity is not affecting the emission footprint of any product(s) that are part of supply chain.
Sustainable development contributions	The project contributes to environmental, social, economic and technological benefits which lead to the sustainable development of the local area and country. PP has mentioned that the PA contributes to 9 SDGs. • SDG 1: No Poverty • SDG 3: Good Health & Well-Being • SDG 4: Quality Education • SDG 5: Gender Equality • SDG 7: Affordable And Clean Energy • SDG 8: Decent Work And Economic Growth • SDG 12; Responsible Consumption And Production • SDG 13: Climate Action • SDG 15: Life On Land CL 01, CL 02 & CL 03 have been raised and resolved.

Additional information relevant to the project	Other Entities involved in the Project: Cropzone Agro Forestry Private Limited (subsidiary of Tramontana Asset Management). It plays a role as the local entity conducting on-ground implementation work to help in the distribution of the ICSs in the local areas. VVB has verified the agreement between the PP and Cropzone Agro Forestry Private Limited /20/. PP has confirmed that the information provided in the PDMR/01/ can be made publicly available. Validation team found that the details, evidence and documents provided by PP are accurate and found acceptable by VVB. The information of the project activity was found acceptable and cross-verified by the validation team.
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3.2 Project Activity Instances in Grouped Projects

The project activity is identified as a grouped project activity. The planned distribution schedule by the PP under different instances is given below.

Instance	Planned ICS Distribution
Region 1	50,000
Region 2	100,000
Region 3	100,000
Region 4	100,000
Region 5	50,000

In the initial phase, a total of 46,160 ICSs were distributed during the first verification. There was only one project activity instance within this verification period.

3.3 Safeguards

3.3.1 Stakeholder Engagement and Consultation

3.3.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	Pre-baseline surveys are conducted across an entire instance region to identify all eligible households. Subsequently multiple beneficiary stakeholder meetings are held to further engage with eligible ICS beneficiaries and convey more detailed information. Additional project stakeholders are directly employed or partnered with at the design and planning phase of a project instance. <u>Primary stakeholders:</u> local families receiving a Tramontana ICS, local employees such as field staff, local administrators (e.g. District Collectors), NGOs <u>Secondary stakeholders:</u> Outside of the local project areas, additional people are employed to support the project activities including manufacturing, transportation etc. Assessment team has checked the attendance sheet /09/ of the people present during the LSC meetings and verified that different categories of the people as mentioned by the PP were present.
Legal or customary tenure/access rights	As part of the project activities, close engagement with the local communities is facilitated to identify any potential concerns and legal conflicts in relation to the project. By means of access to a local project representative or through a designated phone number, stakeholders can raise any concerns or questions they have throughout the project. To date, no conflicts or issues have been raised. The project does not interfere with any legal or customary rights to territories and resources held by project stakeholders. During the onsite visit a few stakeholders were interviewed on this, including confirming whether they have access to a local representative /19/. The PP is also maintaining a grievance logbook /26/ where there were no complaints received till date.
Stakeholder diversity and changes over time	From the attendance sheet/09/ provided by the PP of LSC meetings conducted, there were predominantly socio-economically disadvantaged communities such as Scheduled Tribe/Adivasi or Scheduled Caste/Dalit communities. All the stakeholders are located in India, and it is expected that there will not be any changes over the time.

Expected changes in well-being	The project aims to have a positive impact on stakeholder wellbeing. The social and economic benefits of the project can be outlined as • Reduced drudgery and exposure to risks including wild animal attacks, sexual assault, injuries from tree cutting/wood chopping. • Increased time for skill/economic development • Improved home conditions and quality of life for families • Enhanced safety and reduced burns/injuries for children • Development of IT skills for the local teams employed • Employment opportunities • More hours available for productive income generating activities and schooling. PP has provided the detailed explanation for achieving these targets in the PDMR/01/ which is found to be acceptable.
Location of stakeholders	Most of the stakeholders are the ICS beneficiaries and the remaining are also people from the project region. Therefore, the location of the stakeholders under the first project instance are in ASR district of Andhra Pradesh state in India. The location has been verified during the onsite visit /19/.
Location of resources	Many of the project communities undertake agricultural work on land which has been provided to them (with ownership) by the government. This land is typically near the beneficiary's home. The project does not interfere with such territories of the stakeholders.

3.3.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	All potential beneficiaries of an ICS within a mandal are invited to attend in-person stakeholder meetings, which are advertised in advance through newspapers, text message, pamphlets and/or word of mouth. Every attendee is provided with a physical brochure containing similar information (presented in Figure 10), and also offered refreshments during the meeting.

Consultation outcome	The 21 stakeholder meetings for the first project instance were all successfully conducted, all invited special guests were keen advocates of the project activity, communicating their strong support and highlighting the importance of such an initiative for marginalized and underrepresented communities. PP has provided all questions which were raised during the LSC in the form of detailed meeting minutes, and some of the most common questions in the PDMR/01/. There were no negative comments made by the attendees.
Ongoing communication	People from the local area are employed as field staff and project coordinators. From the onsite visit/19/ it is confirmed that the end users/local stakeholders are in contact with the local coordinators where end users/local stakeholders can raise any comments or concerns to them.
Stakeholder input	The project team diligently documents all minutes, attendance sheets (with attendee signatures) and enquiries raised with corresponding responses during stakeholder meetings, as evidenced by the comprehensive compilation in the PDMR/01/ and supporting documents for each meeting/09/. These documents encapsulate all the topics which were presented during the meetings, details of invited chief guests from the community, and questions raised by the community during stakeholder engagements for the initial project instance. VVB further verifies the information during their site visits through interviews with local stakeholders and validation of supporting evidence presented by the PP/09/. Some recurring questions posed by the local community include: 1. Duration of warranty for the project stoves. 2. Advantages of the project stove compared to traditional cookstoves. 3. Quantity of food that can be cooked on the stove in a single use. These examples of common queries find their responses in Table 6 of the PDMR. Moreover, the Assessment team, during the site visit, obtained stakeholder confirmation of this from the end users and local stakeholders. Consequently, VVB asserts that the stakeholder engagement was conducted transparently and effectively, encompassing detailed discussions on all facets related to the project.

3.3.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	In a positive and comprehensive approach, the project team articulates the advantages of ICS devices and provides a practical demonstration of their safe usage during stakeholder meetings. These sessions serve as a platform for stakeholders to express concerns or seek clarification, with the Project Proponent ensuring that all questions raised and their responses are documented and made accessible in the PDMR/01/ and supporting documents/09/. Furthermore, as part of the ICS distribution process, a formal agreement is executed between the Project Proponent and each beneficiary. The key terms of this agreement are broadly outlined during stakeholder meetings and presented in a more detailed manner at the time of ICS distribution.
Outcome of FPIC discussion	All beneficiaries provided consent to their involvement in the project and the terms of the license agreement/22/, which clearly mentions all the terms in it. The feedback was overwhelmingly positive for project participation based on the stakeholder meetings and pre-baseline surveys conducted. The project has not encroached on land, relocated people without consent, or forced physical or economic displacement. No conflicts have been recorded under the project to date.

3.3.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	The grievance redress procedure was developed based on consultations with stakeholders during the planning stages of the project. The process ensures that any grievances raised by stakeholders are respected, recorded digitally as well as in a grievance logbook and addressed. Additionally, the PP has introduced a cloud-based system which helps the PP to initiate the process of addressing the grievances very soon after the grievance is recorded.
Grievance redress procedure	Any concerns or questions stakeholders may have following stakeholder meetings and during the project life, can be directly

	reported to the local mandal representative or communicated through a dedicated phone number. VVB has confirmed this from the interviews with local stakeholders during onsite visit/19/ and from the grievance logbook provided/26/.
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3.3.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
None	No public comments were received during the public comment period which was from 16-June-2023 to 17-July-2023	-

3.3.2 Respect for Human Rights and Equity

3.3.2.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	As part of the project activities, a training program is implemented for all field staff and coordinators employed. The training enforces strict rules regarding harassment and discrimination. Should any accusations be made by beneficiary families or other villagers, the specific employed individuals will be evaluated and immediately removed from the programme if found to be at fault. Since the onset of the project, no project stakeholders have reported any form of discrimination or sexual harassment. From the interviews with stakeholders during the onsite visit/19/, VVB confirms that there are no complaints on discrimination or sexual harassment from the stakeholders.
Management experience	The Project Proponent has extensive expertise in risk management in financial markets, project development and the management of large teams. Cropzone has strong relationships with the local communities and some experience in on-ground logistics and project implementation.

	Additionally, involvement of digital devices such as Tablets to record the end user's data and managing all data through cloud services makes the assessment team strongly view that the project has a high integrity, and the same has been examined during the field visit/19/.
Gender equity in labor and work	All employees under the project have been paid fairly and in accordance with local customs and workers' rights. Employment of local community members and improving their livelihood and skillset is a central component of the project, particularly given the lack of opportunities available to the marginalised project communities. Where possible the project looks to employ female team members to improve the gender representation that are in work, which is typically low in the project region
Human trafficking, forced labor, and child labor	Human trafficking, forced labour and child labour are strictly forbidden in the project. Background checks are conducted on all staff. A strict 18+ age policy is also followed. No activities causing such things were found from the interviews with the end users and local stakeholders during onsite visits/19/.

3.3.2.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Human rights	The project recognises, respects, and promotes the protection of the rights of all local communities and employees in line with applicable international human rights laws, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples/25/.

3.3.2.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	All project activities are conducted following local customs/traditions, respecting the area's cultural heritage. All stakeholder meetings and surveys are also conducted in the local language and using team members employed from the local community. From interviews conducted with end users during the site visit/19/, the VVB confirms that all the meetings were undertaken and in the local language. PP has also provided photos of the stakeholder

meetings and the newspaper advertisements for the meetings (in the local language) in the PDMR/01/. This makes the assessment team believe that the project meets the criteria defined by VCS.

3.3.2.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Rights to territories and resources	The project does not engage with territory, resource or property rights. The ICS are provided free of cost with a licensing agreement. VVB has cross checked the agreement signed by end users/22/.
Respect for property rights	The project fully respects all property rights and does not impact them through the project activities carried out

3.3.2.5 Benefit Sharing

This section is not applicable as the project does not impact property rights as discussed in Section 3.3.2.4.

3.3.3 Risks to Local Stakeholders and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	PP has identified this risk and provided the multiple preventive measures taken to minimize them in the PDMR/01/, which the VVB has confirmed from the local stakeholder interviews during the site visit/19/.
Working conditions	The Project Proponent (PP) has successfully recognized an enhancement from the baseline scenario, showcasing a positive trajectory. Importantly, the project respects and preserves the established working customs of local communities, contributing to a harmonious integration. With the Improved Cookstove (ICS) introducing cleaner combustion characteristics and heightened thermal efficiency, it not only avoids disrupting traditional practices but significantly enhances the working conditions and livelihoods of end users. This positive impact has been thoroughly verified by the Validation and Verification Body (VVB) through an examination of the monitoring database and end-user interviews conducted during the site visit as documented in /19/.

Safety of women and girls	The Project Proponent (PP) has positively identified a diminished risk compared to the baseline scenario. This reduction is attributed to the significant decrease in fuel collection time resulting from the lowered fuel consumption facilitated by the project activity. The joint validation & verification by the VVB, conducted through desk review and site interviews as documented in /19/, confirms that the primary contributors to fuel collection are females.
Safety of minority and marginalized groups, including children	PP has identified a reduced risk from the baseline scenario. The ICS device has generally been designed to be more compact and safer to ensure that there is minimal risk of burn or injury to users.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	PP has identified a reduced risk from the baseline scenario. The Tramontana ICS significantly reduces toxic indoor air pollution through its improved design, thermal efficiency and cleaner combustion characteristics /15/. From a review of the monitoring database as well as interviews with the end users /19/ during site visits, the VVB observed that there were emissions reductions and minimal local pollution/smoke through usage of the project stove.

3.3.4 Ecosystem Health

No negative environmental impacts were identified for the project. An Environmental Impact Assessment (EIA) is not required for the project as there are only positive environmental outcomes.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	Through the improved design and increased thermal efficiency of the ICS, there is a reduced firewood requirement. This results in reduced deforestation/forest degradation and a positive impact on biodiversity and ecosystems. VVB believes that the implementation of the project leads to a positive impact on local biodiversity.
Soil degradation and soil erosion	The reduced deforestation/forest degradation resulting from the lower firewood requirement leads to less soil degradation and erosion.

Water consumption and stress	The better quality soil conditions as a result of the project activities lead to improved soil water retention. Less water is also needed for cleaning utensils and maintenance of the household owing to the cleaner combustion of the ICS and minimized smoke/soot generation. These points were also highlighted in the monitoring database/24/ by beneficiaries.
Usage of fertilizers	The project activities do not have a measurable impact on the usage of fertilizers.

3.3.4.1 Rare, Threatened, and Endangered species

The project activity is situated in a location that is neither known to be within nor adjacent to habitats inhabited by rare, threatened, or endangered species. Consequently, this section is deemed not applicable to the project.

3.3.4.2 Introduction of Species

N/A

3.3.4.3 Ecosystem conversion

N/A

3.4 Application of Methodology

3.4.1 Title and Reference

The applied methodologies for the project activity are given in table below with their versions. Assessment team has verified the versions of the applied methodologies through the VCS Program details /07/, CDM Methodologies/08/ and the applicability of the methodologies has been explained in section 3.4.2.

Type	Reference ID	Title	Version
Methodology	VMR0006	VMR0006 Energy efficiency and fuel switch measures in thermal applications ¹	1.2

¹ https://verra.org/wp-content/uploads/2023/07/VMR0006-v1.2_PDF.pdf

Methodology	CDM AMS-II G	Energy efficiency measures in thermal applications of non-renewable biomass ²	13.1
CDM Tool	CDM TOOL30	Calculation of the fraction of non-renewable biomass	4.0
CDM Tool	CDM TOOL33	Default values for common parameters	2.0

3.4.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
VMR0006	Project activities must introduce energy efficiency and fuel switch measures in thermal applications (including cookstoves, ovens and dryers) that: - increase thermal efficiency to reduce the consumption of non-renewable biomass, or - switch from fossil fuel (coal or kerosene) to renewable biomass in new or existing improved thermal energy generation units.	The project activity involves the distribution and installation of thermally efficient ICS in rural households in India which use non-renewable biomass. The efficiency of the ICS is significantly more than the traditional cookstoves which has been verified from the thermal efficiency test report/15/. Hence this criterion is applicable.
VMR0006	The project activities must be implemented in households, community-based kitchens, institutions (e.g. schools) or	The project activity is implemented in households of rural areas of the ASR district of Andhra Pradesh in

² <https://cdm.unfccc.int/UserManagement/FileStorage/6TOUCX21D0BHNVRZFWMEKALY94GS7>

	small and medium sized enterprises (SMEs)	the first instance. The assessment has identified the implementation of the project activity from the onsite visit/19/. Hence this criterion is applicable.
CDM AMS-II G	The project cookstove must have a specified rated thermal efficiency of at least 25% as per the manufacturer's specifications, be single-pot or multi-pot and portable or in-situ	The Tramontana ICS installed under the project activity is a high-quality, single-pot portable cookstove, which uses woody biomass as a fuel input in line with manufacturer specifications. The ICS has an efficiency of 36.01%, which has been verified from the thermal efficiency test report/15/. Hence this criterion is applicable.
VMR0006	Project activities characterised as Projects and Large Projects can apply this methodology.	The project activity involves the distribution of 400,000 ICS across India and the estimated annual average GHG emissions reductions generated from this is 1,310,160 tCO ₂ e which is more than 300,000 tCO ₂ e per year. Therefore, the project activity comes under large project as per the VCS standard v4.7/03/. Hence this criterion is applicable.
CDM AMS-II G	The aggregate energy savings of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input	The energy savings for a single project activity is found to be below 60 GWh per year. This is confirmed by the VVB from the calculations in the distribution database spreadsheet /24/ provided by

		the PP, and it is found to be appropriate by the VVB.
CDM AMS-II G	Non-renewable biomass has been used in the project region since 31 st December 1989, using survey methods or referring to published literature, official reports or statistics	VVB has checked table 8 of the document/29/ from UN Food and Agricultural Organization that there was a significant demand for firewood in India observed over the period 1970 to 1995. And VVB also checked the document referred by the PP that over the period 1993 to 1997, 671,000 hectares or equivalently over 1% of total forest cover (dense and open forests) was lost in India. In addition, 1,831,000 hectares of dense forests was either degraded or destroyed according to State of Forestry reports from Forest Survey of India/30/. And for the period from 2001 till 2023, VVB has checked the global forest watch report /28/ where it is mentioned that India lost 2.33Mha of tree cover, equivalent to a 6% decrease in tree cover since 2000. Therefore, it is clear that the use of non-renewable biomass has been in use in the project region since 31 st December 1989. The VVB acknowledges the appropriateness of the data, reinforcing the positive assessment of the project's impact on reducing non-renewable biomass consumption.

CDM AMS-II G	For cases where the biomass is sourced from renewable sources, the project participants should use a corresponding Type I methodology	Not applicable as biomass is not sourced from renewable sources under the project activity
VMR0006	For fuel switch activities, the project must exclusively use renewable biomass and apply the detailed conditions stated in the methodology	This condition is not applicable as the project activity does not involve fuel switching. Instead, the activity involves reducing the consumption of non-renewable biomass by replacing traditional cookstoves with high quality thermally efficient ICS.
CDM AMS-II G	The VCS project document shall explain the proposed method for the distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	Each ICS installed under the project shall be identifiable by the Project Proponent logo and a unique serial number which is displayed on the ICS itself as a QR code and in written form. This is linked to individual users using the Tramontana ICS software application which scans the QR code and records corresponding beneficiary information including personal details, address, date of installation, geolocation of household, user/ID photos etc. The assessment team has verified that all the ICSs distributed have a unique reference number with a QR code both from the distribution database/24/

		and also from the onsite visit/19/.
CDM AMS-II G	The VCS project document shall also explain how the proposed procedures prevent double counting of emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	A standardized license agreement is signed physically (with written signature or thumbprint) by each ICS beneficiary which includes (amongst other conditions) provisions for ownership of all emissions reductions generated by the ICS to be solely with the Project Proponent. The license agreement is generated through the Tramontana ICS software application on a tablet which is then signed by the end user. The assessment team has verified the signed improved cookstoves (ICS) license agreements by the end users from the distribution database/24/.
CDM TOOL30	This tool may be used by: a) DNAs to submit region- or country-specific default fNRB values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or b) project participants to calculate project- or PoA-specific fNRB values.	The VVB has verified that this tool was used to calculate the project specified value for fNRB under option (b) of the applicability conditions. For the PA, the fNRB is calculated as 0.894 in line with equation 1, 2, 3 & 4 of the tool. Assessment of the fNRB is provided under section 3.4.6 of this report. The tool is used by the Project Proponent to calculate project specific fNRB values which has been checked in section 5.4 of the JPD-MR/01/, For future PAs as well, Tool 30 will be applicable. Hence, the criterion is found to be met.

CDM TOOL33	The default values as contained in section 5 of this. tools are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the default values after this date, including in cases where further default values are added to this tool through revisions of this tool before this date.	The project start date is prior to 10th March 2025 and therefore, the default value of 0.15 used from CDM TOOL33 for the efficiency of pre-project cooking devices is valid. Any project activity occurring after 10th March 2025, will use the latest default value available.
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CAR 01 has been raised and resolved.

3.4.3 Project Boundary

The boundary for the grouped project can be represented by the geographical boundary of India. Each project instance will also have its own separate defined boundary within the overall boundary of the grouped project. For the first project instance, this corresponds to the Paderu division of the Alluri Sitarama Raju district of Andhra Pradesh with approximate coordinates of latitude 18.011959 and longitude 82.555030.

The greenhouse gases to be included in the project boundary are presented in the table below.

Source		Gas	Included?	Assessment and conclusion
Baseline	Emissions from the use of non-renewable biomass	CO ₂	Yes	It is considered to be a major source of emissions as a result of the burning of non-renewable biomass in the region. From the baseline database/23/, VVB found the usage of fuel is significant in the baseline scenario.
		CH ₄	Yes	It is considered to be a major source of emissions as a result of the burning of non-renewable biomass in the region. From the baseline

Source		Gas	Included?	Assessment and conclusion
Project				database/23/, VVB found the usage of fuel is significant in the baseline scenario.
		N ₂ O	Yes	It is considered to be a major source of emissions as a result of the burning of non-renewable biomass in the region. From the baseline database/23/, VVB found the usage of fuel is significant in the baseline scenario.
		Other	No	No other sources identified
	Emissions from the use of non-renewable biomass	CO ₂	Yes	From the monitoring database/24/, VVB found the usage of firewood and therefore emissions of this gas have been substantially reduced compared to the baseline scenario.
		CH ₄	Yes	From the monitoring database/24/, VVB found the usage of firewood and therefore emissions of this gas have been substantially reduced compared to the baseline scenario.
		N ₂ O	Yes	From the monitoring database/24/, VVB found the usage of firewood and therefore emissions of this gas have been substantially reduced compared to the baseline scenario.
		Other	No	No other sources identified

3.4.4 Baseline Scenario

The baseline or pre-project scenario is the continued use of woody biomass by local communities to meet their thermal energy needs such as for cooking, heating and boiling water. Use of rudimentary traditional cookstoves is prevalent, most commonly in the form of fixed mud clay stoves or three-stone stoves. This is in line with section 6 of the applied methodology VMRO006 v1.2/07/.

For the first project instance, baselines surveys were conducted across 11 mandals in the Alluri Sitarama Raju district of Andhra Pradesh. This occurred between 3rd November 2022 and 12th December 2022. Pre-baseline surveys were initially conducted to estimate the approximate number of potential ICS beneficiaries in each village. The baseline survey details are verified by the VVB from the baseline database provided by PP/23/.

The procedures for identifying the baseline scenario have been correctly followed and the identified scenario reasonably represents what would have occurred in the absence of the project. VVB confirms that the information provided in the joint PDMR/01/ and the applied methodologies, tools, and documentary evidence are correctly quoted/interpreted and undertaken to a high standard.

CL 04 has been raised and resolved.

3.4.5 Additionality

The project is designed and implemented on a voluntary basis by the Project Proponent and therefore it is a voluntarily coordinated action.

There is no mandated government program or policy in the host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is also not mandated by any law, statute, or other regulatory frameworks, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework.

The project activity installs or distributes the project devices at zero cost to the end user. VVB has verified that the project devices have been distributed free of cost by interviewing the beneficiaries during the onsite visit/19/ and VVB has also verified the signed improved cookstoves (ICS) license agreement where it is clearly mentioned in clause 1a that the PP will not charge anything for the ICS distributed and that it will be distributed free of cost. The agreement has been signed by the beneficiary and the representative of the Cropzone field team conducting the survey. /22/

The project ICS (and any other ICS that may be included in future instances) will be distributed and licenced to beneficiaries free of charge under the project activity, with no sources of revenue generated from the project activity other than from the sale of GHG carbon credits. The project activity is independently undertaken by the Project Proponent on a voluntary basis and is not affiliated with any government schemes or supported by multilateral funds. VVB confirms that the project activity satisfies the applicability conditions as per the section 7 of applied methodology VMR0006 v1.2/07/, therefore the project activity can be deemed as additional.

3.4.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

The project follows methodologies VMR0006 version 1.2/07/ along with AMS-II.G version 13.1/08/. All ex-ante fixed parameters considered for calculation of emission reductions are correctly applied by the PP.

S.No	Parameter	Value applied	Means of assessment
1.	B_{old}	6.77 tonnes	The value of 18.54 kg/day (equivalent to 6.77 tonnes/annum) was derived from direct measurements of daily firewood consumption using digital scales during the baseline survey. These measurements were verified by the VVB during on-site field visits, which included interviews with beneficiaries and direct measurements of firewood consumption in audited households. During these visits, the VVB team also weighed the firewood at the households using a scale. Additionally, all baseline and end users were interviewed about their firewood use in the baseline scenario. The VVB cross-checked the responses and direct measurements

		from the audited samples with the baseline database column "AH", "Fuel used per day (Kg)" and concluded that the value of the parameter is acceptable and appropriate. The project implemented under first PAI is a very specific region like Chintapalle Mandal (Approx 1,100 Meters), Paderu Mandal (Around 900 – 1,200 Meters), GK Veedhi Madal (Around 1,000 Meters), Hukumpeta Mandal (Around 900 – 1,000 Meters), Ananthagiri Mandal (Approx 1,000 meters) etc within Alluri Sitarama Raju district of Andhra Pradesh, at an altitude of around 900 – 1,200 meters above sea level, where low temperatures are reported. Consequently, stoves are used not only for cooking but also for boiling water for bathing and drinking purposes as well as room heating. This multifaceted usage results in higher firewood consumption compared to other regions in India. VVB reviewed few other projects which were registered with more or less same Bold values as tabulated below: <table><tr><th>Sr. No.</th><th>VCS ID</th><th>Bold determined by back calculating By,Saving. (tonnes/yr)</th><th>Date of Registration</th><th>Location</th></tr><tr><td>1.</td><td>3878</td><td>6.244</td><td>20/03/2024</td><td>India (Rajasthan, Gujarat)</td></tr><tr><td>2.</td><td>3250</td><td>6.117</td><td>16/01/2024</td><td>India (Assam)</td></tr><tr><td>3.</td><td>4312</td><td>6.752</td><td>15/02/2024</td><td>Bangladesh</td></tr><tr><td>4.</td><td>4250</td><td>5.222</td><td>08/05/2024</td><td>India (Meghalaya)</td></tr><tr><td>5.</td><td>2380</td><td>7.635</td><td>11/05/2023</td><td>Rwanda</td></tr></table> To cross check the Temperature variations VVB has checked temperature in the particular region from the Climate change Knowledge Portal, World Bank/35/ & Meteologix/36/. Given these conditions VVB concluded that the Bold value observed aligns more closely with the firewood consumption patterns in similar climatic regions, including the project region in Rwanda, despite the geographic differences.	Sr. No.	VCS ID	Bold determined by back calculating By,Saving. (tonnes/yr)	Date of Registration	Location	1.	3878	6.244	20/03/2024	India (Rajasthan, Gujarat)	2.	3250	6.117	16/01/2024	India (Assam)	3.	4312	6.752	15/02/2024	Bangladesh	4.	4250	5.222	08/05/2024	India (Meghalaya)	5.	2380	7.635	11/05/2023	Rwanda
Sr. No.	VCS ID	Bold determined by back calculating By,Saving. (tonnes/yr)	Date of Registration	Location																												
1.	3878	6.244	20/03/2024	India (Rajasthan, Gujarat)																												
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3.	4312	6.752	15/02/2024	Bangladesh																												
4.	4250	5.222	08/05/2024	India (Meghalaya)																												
5.	2380	7.635	11/05/2023	Rwanda																												

			Based on the extensive baseline database, verified measurements, and consideration of local climatic conditions, we find that the Bold value used in the project is adequately justified and aligns with the on-ground realities and specific conditions of the project area.
2.	η_{new_s}	36.01%	This parameter represents Efficiency of the project ICS at the start of the project activity. The value applied has been verified from the manufacturer specifications/18/ and from the efficiency test results/15/.
3.	η_{old}	15%	This parameter represents Efficiency of the baseline cookstove and VVB confirms that the applied 15% efficiency for the baseline stoves is suitable to all the baseline stoves replaced under the project activity. The VVB checked the database of surveys conducted by the PP for the baseline scenario which is showing that all beneficiaries either had a mud clay single/double pot stove or three-stone fixed stove using firewood. According to CDM TOOL33/31/ 5.6 17(a): “For a three-stone fire using firewood (not charcoal), or a cookstove with no improved combustion air supply or flue gas ventilation (i.e. without a grate or a chimney), the default value is 0.15.” Along with this the VVB also checked sources such as IJSET_V3_I12_05.pdf/32/ and Emission factors of wood and charcoal-fired cookstoves – ScienceDirect/33/, where the maximum efficiency of the baseline stoves observed is found to be 15%.
4.	NCV_{wood_fuel}	0.0156 TJ /tonne	This parameter represents net calorific value of the non-renewable woody biomass which has been taken from the applied CDM methodology AMS II.G /08/
5.	EF_{wf,CO_2}	112 tCO ₂ /TJ	This parameter represents the CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario which has been taken from the applied methodology VMR0006/07/.
6.	$EF_{wf,nonCO_2}$	9.46 tCO ₂ /TJ	This parameter represents the Non-CO ₂ emissions factor for the use of non-renewable woody biomass in the baseline scenario which has been taken from the applied methodology VMR0006/07/.
7.	f_{NRB}	0.894	This parameter represents the Fraction of woody biomass saved by the project activity that can be established as non-renewable biomass. Calculated by determining the share of non-renewable woody biomass in the total quantity of woody biomass consumed in the project area. PP has provided a comprehensive fNRB

		calculation sheet/16/ which is based on TOOL 30/10/ and the same has been clearly elaborated in section 5.4 of the JPD-MR/01/ where the value of H has been calculated as per Section 3.1 of the TOOL30, v.04.0/10/ and the value of RB as per Section 3.2 of the TOOL30, v.04.0/10/, which are found to be appropriate. PP has also provided the comparison of the calculated value of fNRB against the values for fNRB reported in relevant scientific literature (including the minimum requirements as per para. 6-a-i of the TOOL30). In line with para 6 of the tool 30 version 4 /10/, which required comparing the fNRB value against published literature, specifically the Bailis et al. (2015)/34/, the fNRB value provided in the main text of the paper is a country-wide average. This encompasses all states of India, many of which have lower fNRB values owing to their low forest cover and firewood usage. The literature referenced in CDM TOOL30/10/, Bailis et al. (2015), reports an fNRB value of 23-24% for India. The PP has provided justification for the difference in their calculated value from this report. As stated in the VCS-PDMR/1/, the differences in these values are attributed to the vintage years used for their calculation and the differing approaches for determining the underlying parameters. This is acceptable to the VVB, as the values in the Bailis Report/34/ are based on data reported in the year 2009, whereas the latest data on forest cover and other underlying parameters are now available. The fNRB value for the first project activity instance was calculated by the PP using data specific to Andra Pradesh, and therefore is a more accurate representation for the project area than a country wide estimate. Additionally, VVB has checked some of the projects registered during 2022 to 2024 under Verra in different states of India which have similar climatic conditions as Andhra Pradesh where the project current project activity is located. The details of the registered projects with their registration date, fNRB values and project location are provided in table below. <table><tr><th>Project ID</th><th>Registration Date</th><th>fNRB Value</th><th>Project Location</th></tr><tr><td>VCS 3240</td><td>15/08/2023</td><td>0.8714</td><td>Odisha</td></tr><tr><td>VCS 3413</td><td>04/01/2024</td><td>0.913</td><td>Maharashtra</td></tr></table>	Project ID	Registration Date	fNRB Value	Project Location	VCS 3240	15/08/2023	0.8714	Odisha	VCS 3413	04/01/2024	0.913	Maharashtra
Project ID	Registration Date	fNRB Value	Project Location											
VCS 3240	15/08/2023	0.8714	Odisha											
VCS 3413	04/01/2024	0.913	Maharashtra											

			<table border="1"> <tr> <td>VCS 3769</td><td>18/12/2023</td><td>0.8755</td><td>Madhya Pradesh</td></tr> <tr> <td>VCS 2664</td><td>23/11/2022</td><td>0.852</td><td>Assam</td></tr> <tr> <td>VCS 3161</td><td>10/01/2023</td><td>0.95</td><td>Telangana</td></tr> </table> Therefore, from the table it is clear that the fNRB, calculated in the project activity is comparable with these projects registered in India. As per Para 13 of the applied Tool 30 version 4/10/ the NRB value is to be crosschecked “with the product of i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr)” when the parameter is determined at the national level. Since, the fNRB and the associated parameters are determined at state level for this grouped project activity this crosscheck is not applicable. In addition to the above checks, VVB has also reviewed the potential of double counting regarding commercial woody biomass consumption. PP has excluded multiple non-domestic wood consumption categories/51/ to keep the fNRB value conservative and to avoid double counting. Further, the domestic wood fuel consumption excludes charcoal consumption as verified from the FAO data/29/ used for the calculation of fNRB. The same has been confirmed by the VVB from the calculation sheet/16/ and hence the VVB confirms that no double counting has been observed. VVB has checked the calculation spreadsheet (which has been provided) for the fNRB steps and found it to be appropriate, and the same has been elaborated in the section 5.4 of the JPD-MR. The calculations use detailed government reports/datasets as inputs, which are uniquely specific to the local region and thereby improving accuracy of the fNRB value selected. CL 07 has been raised and resolved.	VCS 3769	18/12/2023	0.8755	Madhya Pradesh	VCS 2664	23/11/2022	0.852	Assam	VCS 3161	10/01/2023	0.95	Telangana
VCS 3769	18/12/2023	0.8755	Madhya Pradesh												
VCS 2664	23/11/2022	0.852	Assam												
VCS 3161	10/01/2023	0.95	Telangana												
8.	u_d	26%	This parameter represents the Uncertainty deduction for f_{NRB} . The value has been applied as per the applied methodology VMR0006/07/.												
9.	Adj_{LE}	0.95	This parameter represents the Factor to account for leakage related to the non-renewable woody biomass saved by the project activity.												

			The value has been taken from the applied CDM methodology AMS II.G./08/.
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3.4.7 Methodology Deviations

The project activity has not taken any deviations from the applied methodology VMR0006/07/. Hence this section is not applicable.

3.4.8 Monitoring Plan

The project activity involves the distribution of improved cookstoves. A sampling plan is required for the purpose of monitoring and verification of project emissions reductions. A sampling plan has been used by the PP as per CDM Guidelines Sampling and surveys of CDM project activities and programmes of activities/11/.

As per CDM Standard Sampling and surveys for CDM project activities and programmes of activities/12/, a 90/10 confidence/precision is to be used for all small-scale projects and 95/10 for all large-scale projects, and in line with methodology AMS-II G/08/ a 90/10 confidence/precision when undertaking annual sampling and 95/10 when adopting biennial sampling inspections. PP has applied a minimum 95/10 confidence/precision when sampling the parameters for monitoring purposes.

Parameter	Description	Monitoring frequency	Monitoring methods to be applied
$n_{y,i,j}$	Proportion of project devices of type i in instance j operating during year y	Expected annually and will occur at least biennially	Site visits to undertake visual examination and conducting detailed surveys/interviews
μ_y	Stove usage adjustment factor accounting for any use of pre-project or other stoves during year y	Expected annually and will occur at least biennially	Site visits to undertake visual examination and conducting detailed surveys/interviews
$\eta_{new,y,i}$	Thermal efficiency of ICS of type i in year y	Expected annually	WBTs conducted by independent technical laboratories

3.5 Non-Permanence Risk Analysis

This is not applicable for the proposed VCS project as it is a Non AFOLU project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

46,160 ICS have been distributed during the first monitoring period, which is from 21/01/2023 to 23/08/2023, in Andhra Pradesh in India under the first instance of the project activity.

Implementation Status	Assessment steps, evidence checked, & conclusion:
Project implementation	46,160 ICSs have been distributed during the first monitoring period which is from 21/01/2023 to 23/08/2023 in the ASR district of Andhra Pradesh in India as a part of the first project activity instance (PAI). VVB has verified this from the extensive distribution database/24/. The JPD-MR clearly indicates that the project activity instances are defined in terms of regions and not states as one state may have multiple instances given its large size.
Monitoring plan	As detailed in section 3.4.8, the Project Proponent (PP) has actively executed the monitoring plan. The Validation and Verification Body (VVB) has thoroughly verified, through on-site visits and interviews/19/, that the local coordinators consistently conduct regular visits to check ICS use and gain stakeholder feedback. These visits are integral to ensuring the ongoing operational status of the Improved Cookstoves (ICSs), as documented in /14/.
AFOLU-specific project implementation	N/A

4.2 Accuracy of Reduction and Removal Calculations

Data and Parameters monitored:

S.No.	Parameter	Monitoring Frequency	Means of assessment
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1.	$n_{y,i,j}$	At least annually	During the site visit VVB confirms that 100% of the beneficiaries said that the ICSs are operational and there was no breakdown so far. VVB has verified this from the site visit/19/ and the grievance log maintained by the PP/26/. As of now there are no complaints on ICS and feedback is positive.
2.	$N_{0,i,j}$	At the time of commissioning/distribution of ICS	During the first monitoring period 46,160 ICSs were distributed in the project region which was verified by the VVB from the distribution database/24/. Additionally, the VVB has crosschecked the manufacturer sale purchase agreement/20/, and confirmed the same.
3.	n_{HH}	At the time of commissioning/distribution of ICS	Over the first monitoring period one ICS has been distributed to each household. VVB has verified this from the onsite visit/19/ and from the distribution database/24/ as well. The assessment team has confirmed that some of the end users share common names. However, after reviewing the Aadhar/ID card and BPL Card provided respectively in columns F and G of the distribution database spreadsheet, it was confirmed that there is no double counting or repetitive individuals reported.
4.	μ_y	Expected annually but at least biennially	Based on the monitoring survey sample results for the first period, it was observed and explicitly confirmed by all beneficiaries that their pre-project stove had been removed or demolished, and also confirmed that no one was using

			their original stove. VVB has also verified this from the Improved cookstove licence agreement signed by each household/22/ and from the site visit. CL 05 has been raised and resolved.
5.	$\eta_{new,y,i}$	Annually	The thermal efficiency of the project stove was given as 36.01%, which has been verified from the manufacturer specifications/18/ and the thermal efficiency test report/15/.
6.	Life Span	Once at project installation	It is considered that the project device may be in operation beyond 7 years with proper maintenance and this will be checked and reported during monitoring.
7.	Date of commissioning of project device i and the corresponding instance j	Once at ICS installation	The date of commissioning of each project device is being maintained in the distribution database/24/ by the PP.

4.3 Quality of Evidence to Determine Reductions and Removals

The assessment team affirms the authenticity of the calculations and data, attesting to their high quality and integrity. The supporting documents submitted for validation and verification not only meet the required standards but go above and beyond in exhibiting information transparency and traceability.

During the field visit, the assessment team meticulously assessed the quality and extensiveness of the data given in the databases and supporting documents to validate its authenticity. The calculations embedded in the databases and documents were scrutinized to ensure accuracy, instilling a high level of confidence in the project following completion of the assessment process.

The assessment team attests to the strong impact of the project on the UN Sustainable Development Goals (SDGs), reflecting an alignment with its objectives. The evidence gathered

for the entire monitoring period is not only comprehensive but also verifiable, demonstrating that the data collection system adheres to the monitoring plan and applied methodologies.

Furthermore, the assessment team expresses satisfaction with the quality of evidence utilized to determine greenhouse gas (GHG) reductions, affirming it to be of a high calibre. The Joint PDMR/01/ also provides detailed information regarding all roles and responsibilities, contributing to an understanding of the project's governance and management structure.

5 VALIDATION AND VERIFICATION OPINION

5.1 Validation and Verification Summary

Earthood Services Private Limited (Earthood) performs the joint validation and verification of the emission reductions for the VCS project activity “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID- 4457) for the crediting period 21/01/2023 to 20/01/2030 and monitoring period 21/01/2023 to 23/08/2023. Tramontana Impact Limited is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project activity. It is our responsibility as VVB to express an independent validation and verification statement on the reported GHG emission reductions from the project activity.

Earthood commenced the joint validation and verification based on the baseline and monitoring methodologies VMR0006 version 1.2 /07/and CDM AMS-II G version 13.1/08/, the monitoring plan contained in the Joint PD and MR Version 3.0 /01/and VCS Program Guidelines version 4.4/02/ as per the process described under Section 2 of this report.

The VVB has carried out this joint validation and verification with a reasonable level of assurance as stated in section 1.3 above. Further, the verification of the GHG statement was conducted in accordance with ISO 14064-3.

Overall upon completing our review as VVB, we confirm that:

- The project activity consistently demonstrates high standards of quality and integrity, and ensures transparency and traceability in all aspects. The extensive impact of the project is also observed from the adherence to 9 Sustainable Development Goals (SDGs).
- The project activity aligns with all relevant host country criteria and adheres to VCS rules and requirements. Likewise, it complies comprehensively with all conditions outlined in the latest version of the specific applied methodologies.

- The consultation with local stakeholders is conducted extensively, and follows both host country and VCS requirements, affirming a commitment to inclusivity and strong community engagement. Note that an additional environmental impact assessment was not conducted as there is no risk identified relating to soil degradation, impact on biodiversity and ecosystem, water consumption and stress & usage of fertilizers.
- The monitoring plan stands out for its robust structure, ensuring a comprehensive mechanism for overseeing project progress. All detailed information presented in the VCS PDMR/01/ has been consistently applied so far, attesting to the project's commitment to quality.
- The implementation of the project aligns with the description outlined in the VCS PDMR/01/, underscoring the project's commitment to fulfilling its proposed objectives. Moreover, the Project Proponent (PP) has taken a significant stride in embracing digital technologies, such as the development of their own proprietary app and use of Tablets for on-site data recording. This innovation has marked a substantial leap in enhancing project quality and data accuracy.
- The project activity ensures the uniqueness of all the ICS distributed through the use of a proprietary app, photos of users with their ICS etc. The assessment team confirmed that there was no double counting or repetitive households reported. The site visit conducted by the VVB was extensively done with a large area covered to ensure accuracy. At this time it was also possible for the VVB to verify the coordinates of each household interviewed due to geographical mapping of the end users undertaken by the PP. The VVB also cross checked the Aadhar cards of the end users during their onsite visit/19/ and found them to be appropriate with the distribution database/24/.

Altogether, the VVB is of the opinion that the project activity is unique and of a very high standard in comparison to related projects owing to the applied approaches by the PP including: data acquisition through an app, household geographical mapping, photos of users and documents, extensiveness of stakeholder engagement, order of magnitude larger sampling and heavy duty ICS used.

5.2 Validation Conclusion

Earthood has performed the validation by obtaining evidence, conducting a comprehensive site visit, and requesting additional information. The Project Proponent (PP) has undertaken a detailed calculation of estimated emission reductions in accordance with the applied methodology VMR0006 version 1.2/07/. The provided explanations by the PP are deemed necessary by Earthood to give a strong level of assurance that the projected GHG emissions reductions are not only attainable but will, in fact, be achieved. This is exemplified by the commitments to high standards of accuracy and integrity shown throughout the validation process.

Crediting Period: From 21-January-2023 to 20-January-2030

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21-Jan-2023 to 31-Dec-2023	206,521	—	—	206,521	—	206,521
01-Jan-2024 to 31-Dec 2024	860,231	—	—	860,231	—	860,231
01-Jan-2025 to 31-Dec 2025	1,646,313	—	—	1,646,313	—	1,646,313
01-Jan-2026 to 31-Dec 2026	1,719,647	—	—	1,719,647	—	1,719,647
01-Jan-2027 to 31-Dec 2027	1,641,975	—	—	1,641,975	—	1,641,975
01-Jan-2028 to 31-Dec 2028	1,555,844	—	—	1,555,844	—	1,555,844
01-Jan-2029 to 31-Dec 2029	1,459,791	—	—	1,459,791	—	1,459,791
01-Jan-2030 to 20-Jan-2030	80,801	—	—	80,801	—	80,801
Total	9,171,123	—	—	9,171,123	—	9,171,123

CL 08 has been raised and resolved.

5.3 Verification conclusion

Earthood has executed the verification process involving obtaining evidence, conducting a comprehensive site visit, and collating additional information to establish a robust assessment.

The estimated emissions reductions for the specified monitoring period stand at 79,589 tCO₂e. Notably, this figure surpasses the achieved emission reductions of 62,743 tCO₂e, as verified from the Emission Reduction (ER) sheet documented in /13/. This discrepancy, where emissions are lower than estimated, eliminates the need for further assessment, and reflects a sufficient level of accuracy in the initial projections.

This overall outcome from verification instils a high level of confidence in the project's impact on reducing greenhouse gas emissions and achieving the Sustainable Development Goals (SDGs).

Verification Period: From 21-January-2023 to 23-August-2023

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-January-2023 to 23-August-2023	62,743	-	-	62,743	-	62,743
Total	62,743	-	-	62,743	-	62,743

5.4 Ex-ante vs Ex-post ER Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21-January-2023 to 23-August-2023	79,589	62,743	21.2%	The variation in emissions reductions is attributed to factors such as the monsoon weather between June and August, causing some delays in the distribution of Improved Cookstoves (ICS). Furthermore, additional time was invested in conducting resurveys due to observed errors in the database data recorded by the field team. It's worth noting that these resurveys were essential to meeting the pre-defined high standards set by the Project Proponent, and as the team gains experience, the need for resurveys is expected to decrease.

				During the first monitoring period, a total of 46,160 ICS were distributed, which is slightly below the estimated figure of 58,630 by this stage. Despite these variations, the project remains on a positive trajectory, and the experience gained by the team is anticipated to contribute to further efficiency and accuracy in future distribution efforts.
Total	79,589	62,743	21.2%	

Approved by



Dr. Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 16-07-2024

Place: Gurugram, Haryana

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

The project activity does not involve any sensitive information. Hence the section is not applicable.

Section	Information	Justification	Assessment method conclusion
NA	NA	NA	NA

APPENDIX 2: LIST OF DOCUMENTS REFERRED

S.N o.	Title of Document	Version	Date
1.	VCS PDMR	1.5	16-07-2024
2.	VCS Program guide	4.4	29-08-2023
3.	VCS Standard	4.7	16-04-2024
4.	VCS VVB Manual	3.2	19-10-2016
5.	No double counting Declaration	1.0	30-10-2023
6.	Employee Records	1.0	02-11-2023
7.	VMR0006: Energy efficiency and fuel switch measures in thermal applications	1.2	06-07-2023
8.	CDM AMS-II G: Energy efficiency measures in thermal applications of non-renewable biomass	13.1	05-03-2024
9.	Local stakeholder meetings	1	14-11-2022 to 05-12-2022
10.	Tool 30 Calculation of the fraction of non-renewable biomass	4.0	08-09-2022
11.	CDM Guidelines Sampling and surveys of CDM project activities and programmes of activities	9.0	13-09-2012
12.	CDM Standard Sampling and surveys for CDM project activities and programmes of activities	9.0	13-09-2012
13.	Emission Reduction calculation sheet	1	23-11-2023
14.	Monitoring + Sampling database	1	22-01-2023 to 23-08-2023
15.	Third party testing certificates	1	17-01-2020
16.	fNRB calculation sheet	-	-
17.	Project webpage	-	-

18.	Manufacturer's brochure	-	2022 - 23
19.	Onsite visit documents	-	15-10-2023 to 17-10-2023
20.	Executed sale purchase agreement	-	21-09-2022
21.	Ownership Rights	-	21-01-2023 to 23-08-2023
22.	End user agreements	-	21-01-2023 to 23-08-2023
23.	Baseline Database	-	03-11-2022 to 12-12-2022
24.	Project distribution database	-	21-01-2023 to 23-08-2023
25.	https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO::P12100_ILO_CODE:C169	-	-
26.	Grievance logbook	-	-
27.	https://weatherandclimate.com/india/andhra-pradesh#:~:text=Andhra%20Pradesh%2C%20India%20Climate&text=Located%20at%20an%20elevation%20of,4.66%25%20higher%20than%20India's%20averages.	-	-
28.	Global forest watch report for the period from 2001 to 2023 https://www.globalforestwatch.org/dashboards/country/IND/?category=forest-change&location=WYJjb3VudHJ5IiwSU5EI0%3D	-	-
29.	https://www.fao.org/forestry/statistics/80570/en	-	-
30.	Forestry reports from Forest Survey of India https://fsi.nic.in/index.php	-	-
31.	TOOL 33: Default values for common parameters	2.0	-

32.	Thermal Efficiency of Wood Burning Traditional Mud Cookstoves and Their Impact on Indoor Air Quality in North-Western Himalayas https://ijiset.com/vol3/v3s12/IJISSET_V3_I12_05.pdf	-	-
33	Emission factors of wood and charcoal-fired cookstoves https://www.sciencedirect.com/science/article/abs/pii/	-	-
34	Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional wood fuels. Nature Climate Change, 5(3), https://www.researchgate.net/publication/271503594_The_Carbon_Footprint_of_Traditional_Woodfuels .	2015	Others
35	Climate Change Knowledge Portal, World Bank: https://climateknowledgeportal.worldbank.org/country/india	-	Others
36	Meteologix: https://meteologix.com/in/observations/andhra-pradesh/temperature/20231016-0700z.html	-	Others

APPENDIX 3: ABBREVIATIONS

Abbreviations	Full texts
BE	Baseline Emission
BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CO2	Carbon dioxide
CM	Combined Margin
CP	Crediting Period
DR	Desk Review
EB	Executive Board
EF	Emission Factor
ER	External Resource
EIA	Environmental Impact Assessment
Earthood	Earthood Services Private Limited
FAR	Forward Action Request
GHG	Green House Gas
GP	Grouped Project Activity
IRR	Internal Rate of Return
MoU	Memorandum of Understanding
MWh	Mega Watt Hour
MR	Monitoring Report
PAI	Project Activity Instance
PDMR	Project Description & Monitoring Report
PP	Project Proponent

tCO ₂ e	Tonnes of Carbon dioxide equivalent
TA	Technical Area
UNFCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit
VVB	Validation and Verification Body

APPENDIX 4: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Name	Sushant Vashisht		
Education	M.Sc. Environmental science and Technology		
Experience	1+ years		
Field	Environment science and technology		
Approved Roles			
Team Leader	YES (VM)		
Validator	YES (VM)		
Verifier	YES (VM)		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	YES (VM 1.2, 3.1)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	23/06/2023
Approved by	Deepika Mahala (Technical Manager)	Date	23/06/2023

Competence Statement			
Name	Shifali Guleria		
Education	M.Sc. (Environmental Studies and Resource Management), TERI University		
Experience	3+ year		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	YES (AMS-I.A., AMS-II.G., AMS-II.E., AMS-III.A.V., AMS-I.D, ACM0002)		
Local expert	YES		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (1.2, 3.1)		
Reviewed by	Deepika Mahala	Date	18/02/2022
Approved by	Ashok Gautam	Date	18/02/2022

Competence Statement			
Name	Nikhitha Chinthala		
Education	M.Tech (Renewable Energy) B.Tech (Electrical and Electronics Engineering)		
Experience	-		
Field	Renewable Energy		
Approved Roles			
Team Leader	NO		
Validator	NO		
Verifier	NO		
Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Trainee	YES		
Reviewed by	Shifali Guleria (Quality Manager)	Date	12/07/2023
Approved by	Deepika Mahala (Technical Manager)	Date	12/07/2023

Competence Statement			
Name	Shreya Garg		
Country	India		
Education	M.Sc. (Climate Science & Policy), TERI University		
Experience	9 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., AMS.III.BL, ACM0002, ACM0012		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2, TA 3.1)		
Reviewed by	Shifali Guleria	Date	26/04/2022
Approved by	Deepika Mahala	Date	26/04/2022
Approved by	Deepika Mahala (Technical Manager)	Date	12/07/2023

Competence Statement	
Name	Ashok Gautam
Country	India
Education	M. Sc. (Environmental Sciences) M. Tech. (Energy & Environmental Management)
Experience	16 Years +
Field	Energy, Climate Change & Environment
Approved Roles	
Team Leader	YES
Validator	YES
Verifier	YES

Methodology Expert	AMS-I.D., AMS-I.A., AMS-I.C., AMS-I.E, AMS-II.D., AMS-II.G., AMS-III.E., AMS-III.H., AMS-III.Q, AMS-III.Z., AMS-III.AV., AMS III.AR, AM0029, AM0025, AM0056, ACM0001, ACM0002, ACM0004, ACM0012, ACM0006, AM0018, ACM0017, ACM0009, AM0034, AMS.I.B, ACM0016, AMS-III.BL, AMS-II.L, AMS-I.I., AMS-III.A.O., ACM0010, ACM0025		
Local expert	YES (India)		
Financial Expert	YES		
Technical Reviewer	YES		
TA Expert	YES (TA 1.1, TA 1.2, TA 3.1, TA 13.1)		
Reviewed by	Shifali Guleria	Date	06/03/2023
Approved by	Deepika Mahala	Date	06/03/2023

APPENDIX 5: LIST OF FINDINGS

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	NA	Section no.	NA	Date : DD/MM/YYYY
Description of FAR				
Project participant response				Date : DD/MM/YYYY
Documentation provided by project participant				
VVB assessment				Date: DD/MM/YYYY

Table 1.CL from this validation

CL ID	01	Section no.		Date : 06/10/2023
Description of CL				
For SDG 01, PP shall clarify how the project activity which is an ICS distribution qualifies for the indicators 1.2.1 which specifies applicability for “Proportion of population living below the national poverty line, by sex and age “and 1.4.1 for “Proportion of the population living in households with access to basic services”. With respect to the baseline survey conducted by PP, the source of the fuel i.e., wood logs are being gathered from the nearby forest which is available for free in nature. PP is requested to clarify how the project activity is being qualified for SDG indicator 1.2.1.				
Project participant response				Date : 25/10/2023
This is detailed in Table 5. It is observed that 100% of beneficiaries in the first project instance own a Below Poverty Line (BPL)/ration card issued by the state government, signifying that the project communities live below the poverty line. Through the improved stove efficiency of the distributed ICS, the volume of firewood consumed by each household is reduced, leading to time savings in firewood sourcing/collection. Consequently, 100% of respondents from monitoring surveys were found to use the time savings to contribute to the main family income, thereby helping to reduce poverty levels and achieving SDG 1.2.1. Through the provision of ICS to households, families are now provided with access to the basic service of clean cooking in line with SDG 1.4.1.				
Documentation provided by project participant				
VVB Assessment				Date: 06/12/2023

The assessment team has verified that the intended beneficiaries come from rural areas in India and fall under the poverty line (BPL) category, as defined by the Government of India. The Validation and Verification Body (VVB) has substantiated this information through on-site visits and database examination, cross-referencing it with the BPL cards issued by the Government of India. It is also confirmed from the database that all beneficiaries were found to use the time savings to help with the main family work. Consequently, CL#01 is considered closed, affirming the positive impact of the project on the targeted population and their livelihoods.

CL ID	02	Section no.	Date : 06/10/2023
Description of CL			
For SDG 03, PP shall mention if there is any chimney present and clarify how the project activity qualifies for the indicator 3.9.1 which specifies “Mortality rate attributed to household and ambient air pollution” as in project activity is causing for some amount of smoke/ soot generation as per the ‘Contribution of Project Activity and Monitoring Approach’ column of table 4 in the PDMR.			
Project participant response			Date : 25/10/2023
From baseline survey results for the first project instance, it was observed that 85.9% of respondents were using their traditional stove in an enclosed space. Further, 48.7% had no provisions for smoke removal, 33.2% had some small opening in the walls or ceiling, 18% had a window, and only 0.1% indicated having a chimney. These details have been added into Tables 4 and 5 of the PDMR.			
Localised household smoke pollution is estimated to have contributed to over 800,000 deaths in India in 2019 (see page 4 reference of PDMR). As presented in Table 5, monitoring survey results indicate that 100% of beneficiaries stated that localised smoke pollution had decreased by using the Tramontana ICS. In addition, the portable nature of the ICS has given flexibility to beneficiaries to use their ICS outdoors. As a consequence 100% of families stated experiencing improvements to their health. This leads to reduced mortality rates, satisfying SDG indicator 3.9.1.			
Documentation provided by project participant			
VVB Assessment			Date: 06/12/2023
The assessment team, through on-site verification, has affirmed that all sampled households utilize their portable stove. Interviews with end users during the VVB site visit have provided confirmation that the project stove emits minimal smoke compared to traditional stoves, resulting in improved health conditions including reduced eye irritation and coughing which were stated. This positive feedback directly from end users strengthens the assessment team's belief in the efficiency of the project stove and its notable enhancement to the daily cooking conditions of the end users. As a result, this finding is considered closed.			

CL ID	03	Section no.	Date : 06/10/2023
Description of CL			

For SDG 04, PP shall clarify how the project activity which is an ICS distribution qualifies for the indicators 4.3.1 which specifies applicability for “Participation rate of youth and adults in formal and non-formal education and training in the previous 12 months, by sex “and 4.4.1 for “Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill”. As mentioned in PDMR “The impact on this SDG can be assessed from the number children involved in stove related tasks in the baseline, and then through the project activity, the number of beneficiaries (predominantly children) that are able to spend more time in education.”, however as per the baseline survey sheet provided, the fuel is being collected by different family members including adult daughters, adult sons, spouse, self. PP is required to include proper justification by providing the evidence to substantiate the children/ percentage of children involved in stove related tasks.

Project participant response

Date : 25/10/2023

As discussed in Table 5, the project directly employs local people (33 full time and 268 part time) for the project activities who are given extensive training for their roles, including IT skills (using the Tramontana application) and data acquisition methods to maintain high quality/integrity.

From monitoring surveys it is identified that 66% of respondents indicated that their family (primarily children) were already able to spend more time in education as a result of the time savings from using the Tramontana ICS.

Thus both indicators 4.3.1 and 4.4.1 are positively impacted.

We find that 29% of children are currently involved in stove usage. However, we do not have specific information on their additional involvement in other stove related tasks such as fuel collection. As a result, we have modified the wording in Table 4 accordingly.

From the second project instance onwards we are planning to update the baseline surveys to include an explicit question on whether children are involved in fuel related tasks e.g. collection, cutting/drying.

Documentation provided by project participant

VVB Assessment

Date: 06/12/2023

Through on-site interviews with end users, it was confirmed that the project stove has had a positive impact by significantly reducing their daily wood collection time, on average, from a daily to a weekly basis. This positive change allows them to allocate more time to their children, fostering increased attention to their education, as confirmed by the monitoring database. As a result, the Verification Body (VVB) has concluded that the SDG indicator 4.3.1 has been successfully met, leading to the closure of CL#03.

CL ID	04	Section no.	4.1 of PDMR	Date : 06/10/2023
Description of CL				

After the distribution of project devices to the end-users, the baseline device (conventional three-stone stove) may still be used by some of the end-users. PP is requested to clarify the following:	
1. What are the measures taken for the discontinued use of traditional stove. 2. In case of any continued use of traditional stove, how the factor will be taken into emission reduction calculation.	
Project participant response	Date : 25/10/2023
1. As highlighted in section 1.14 of the PDMR, there is a requirement for the demolition or removal of pre-project stoves in households prior to receipt of a Tramontana ICS. Further, all ICS recipients sign an end-user agreement when receiving an ICS confirming they accept this condition and that they shall not use any other cookstoves going forward. The signed agreements can be viewed in the Distribution database spreadsheet provided. 2. If for any reason, it is discovered during monitoring that a recipient is using a non-project stove, this is accounted for through the parameter μ_y using the calculation outlined in section 6.2 on pages 92 - 93 of the PDMR.	
Documentation provided by project participant	
VVB Assessment	Date: 06/12/2023
1. PP has provided the improved cookstove license agreement signed by each end user, where the end users have accepted for the demolishing of their baseline stove. Hence, the findings are closed. 2. PP has provided the detailed calculation in the PDMR. Hence, the finding is closed.	

CL ID	05	Section no.	4 of PDMR	Date : 06/10/2023
Description of CL				
PP is requested to clarify how the consumption of wood in accordance with the seasonal variation will be accounted for. Request to update the VCS PD accordingly.				
Project participant response				Date : 25/10/2023
ICS beneficiaries are surveyed based on their average fuel consumption over the year, accounting for any minor seasonal variations. The usage of an average has been reiterated in section 3.4 describing the baseline scenario.				
Documentation provided by project participant				
VVB Assessment				Date: 06/12/2023
VVB confirms that the consumption of wood has been accounted as the average fuel consumption over the year which has been confirmed from the baseline survey sheet provided by the PP. Same has been reflected in the revised PDMR. Hence, CL#05 is closed.				

CL ID	06	Section no.	1.12 of PDMR	Date : 17/11/2023
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Description of CL	
1. In the manufacturer brochure provided by the PP it is showing the weight of the cookstove as 11kg, however PDMR is showing 10.1 kg. PP is requested to clarify if there is modification conducted in the project stove before the distribution. 2. PP is requested to clarify how the life span of the cookstove is taken as 7 years as there are no details mentioned in the manufacturer brochure provided.	
Project participant response	Date : 20/11/2023
1. No modification to the stove has been made prior to distribution. The 11kg value in the brochure is an estimation by the manufacturer and also includes the packaging materials to transport the ICS. The PP has re-measured the mass of the ICS on its own using an electronic scale and obtained a recording of 9.471 kg. The PDMR has been updated to reflect this value. 2. The average expected ICS lifespan of 7 years is based on the manufacturer's experience and expertise from many years of stove development. In any case, the value is not used for emissions reductions calculations, and the PP is replacing and repairing all ICS that breakdown or malfunction during its lifetime.	
Documentation provided by project participant	
VVB Assessment	Date: 06/12/2023
1. PP has revised the weight of the ICS. Hence the finding is closed. 2. PP has provided a sufficient explanation on the life span. Hence the finding is closed.	

CL ID	07	Section no.	fNRB Calculation sheet	Date : 17/11/2023
Description of CL				
1. The total of VDF+MDF+OF is 29,137.40. PP is requested to clarify the inconsistency. 2. As per the source, the value of F_{other} is 8,254.84 sq/km. PP is requested to clarify the inconsistency. 3. According to the referred census data 2011, the total household size (rural and urban) of Vishakhapatnam is 334,898. However, 793,161 households using firewood are mentioned in the sheet which is greater than the total population reported in the census data of the district. PP is requested to clarify how the household size has been calculated in the sheet.				
Project participant response				Date : 20/11/2023

1. In CDM TOOL30, F_{forest} is defined as the extent of forest in hectares. From the source used (India State of Forest Report 2019), there are two potential values for this parameter: the “Recorded Forest Area (RFA)” in paragraph 2 of page 3, and the “Total Forest Cover” in table 11.1.2 (i.e. VDF+MDF+OF). We have chosen to use the RFA for two reasons: • RFA is larger than the Total Forest Cover and therefore produces a more conservative f_{NRB} value (0.894 as compared to 0.955) • The RFA is the official value for 2019 from government records (under the Indian Forest Act). The Total Forest Cover is based on older satellite interpretation (2017-2018) and does not include scrub forest which is a forest type.	
2. In CDM TOOL30, F_{other} is defined as the extent of wooded land not included in the state-recognised forest area. The value used, 8,932 sq km, is the “Extent of TOF” from table 11.1.14 of the India State of Forest Report 2019. Extent of TOF (Trees Outside Forests) is defined as the sum of forest cover outside of the RFA plus Tree Cover (defined as forest with less than 10% density). • The value stated in your question, 8,254.84 sq/km, relates to scrub forest (table 11.1.2 of the India State of Forest Report 2019). We do not think this is the correct value to use, as scrub forest is a component of the forest and is included in the RFA value to determine the F_{forest} parameter.	
3. From the 2011 census report “A-01: Number of villages, towns, households, population and area (India, states/UTs, districts and Sub-districts)”, Visakhapatnam has a total number of 1,097,042 households (refer to cell J107). • From the 2011 census report “HL-14: Percentage of households to total households by amenities and assets, Andhra Pradesh, district Visakhapatnam”, 72.3% of households use firewood for cooking. This value was calculated by taking the average over all subdistricts of Visakhapatnam. • Multiplying the two values above together we obtain the number of households consuming firewood for cooking within the Visakhapatnam district, $1,097,042 * 72.3\% = 793,161$ • 793,161 is less than the total number of households in Visakhapatnam 1,097,042 • It is unclear where the value 334,898 in the question has been taken from	
Documentation provided by project participant	
VVB Assessment	
Date: 06/12/2023	

1. VVB confirms that the data provided in the PDMR is consistent with the referred data. The values as mentioned by the PP are in line with the referred documents. The documents have been reviewed by the VVB and are found to be acceptable. Closed.
2. The justification provided by the PP is found to be acceptable by the VVB after a review of the referred documents. Hence, the finding is closed.
3. The approach applied by the VVB to calculate the population is found to be acceptable by the VVB. Closed.

CL#07 is closed.

CL ID	08	Section no.	Ex-ante ER sheet	Date : 17/11/2023
Description of CL				
1. The average value of baseline fuel consumption per day is 18.54 kg. However, emission from 1kg Wood lies between 1.6 - 1.85 kg (dependent on species). PP is requested to clarify how it insured that the Bold is calculated appropriately. Kindly refer to the document link provided below. https://cdm.unfccc.int/Panels/ssc_wg/SSCWG05_repan_07_Scenario_NRB2RB.pdf 2. PP is requested to clarify how the efficiency is being determined as 3%/year. 3. In line with the AMS II.G, para 37a, "The stove efficiency decreases linearly over time, i.e. at a constant rate which is equal to the difference between the initial and final efficiencies divided by the lifespan of the project device". PP is requested to clarify how the formula has been applied in line with the methodology. 4. Ex-ante Emission Reductions provided on the Verra page and in the PDMR are not matching. PP is requested to make them consistent.				
Project participant response				Date : 20/11/2023

1. B_{old} is defined as the annual quantity of woody biomass which would have been used in the household in the absence of the project activity in tonnes/person/year, as per the VMR0006 methodology. The emissions factor which the question refers to is not involved in the calculation of B_{old} .
 - To ensure B_{old} is calculated accurately, a new value is calculated for each project instance region and a large sample size is taken. For the first project instance, the baseline sample size was 1,793 households (approx. 3% of total households) which was then averaged.
 - To convert B_{old} into CO₂-equivalent emissions, we use the net calorific value of non-renewable woody biomass (NCV) and the emission factors (EF_{Non_CO2} and EF_{CO2}), see Equation 1 on page 74 of the PDMR. The values used for NCV and EF are default values provided by the IPCC, in line with the VMR0006 methodology. The default values represent averages and incorporate any variations that may be observed across a range of wood types. The value of 1.6 - 1.85 kg stated in your question is not in line with the Verra methodology used for the project.
2. Firstly please note that this value is purely for ex-ante calculations to estimate emissions reductions. The stove efficiency for actual emissions reductions is measured on an annual basis using water boiling tests, see pages 91-92 of the PDMR.
 - To determine the ex-ante linear efficiency reduction value, the methodology outlined in paragraph 37 (a) of AMS-II G is applied. This involves taking the difference between the initial and final efficiencies, and dividing by the lifespan of the project device. Using the initial efficiency of 36.01% (see Table 2 of PDMR), a terminal efficiency of 20% (following AMS-II G guidelines) and a life span of 7 years, an annual efficiency reduction of 2.29% is obtained. We then select a more conservative value of 3%. This text has been added into the PDMR on page 78.
3. See response to question 2
4. The Verra registry page was updated at the time of listing in May 2023 when the draft version 1.2 of the VMR0006 methodology was available. We have since applied the finalised version 1.2 of the methodology issued in July 2023 and have stated the values in the PDMR. We will communicate with Verra at the time of submission and request for the values in the registry page to be updated accordingly.

Documentation provided by project participant

VVB Assessment

Date: 06/12/2023

1. The justification provided by the VVB is found to be acceptable by the VVB after a review of the referred documents. Hence, the finding is closed. 2. The justification provided by the VVB is found to be acceptable by the VVB after a review of the revised section 5.4 of the PDMR. Hence, the finding is closed. 3. The justification provided by the VVB is found to be acceptable by the VVB after a review of the revised section 5.4 of the PDMR. Hence, the finding is closed. 4. The ex-ante Emission Reductions reported in the Verra page was from the very initial phase of project i.e., listing. The estimated ERs are to be made consistent between the project documents and project activity web page. Hence, point 4 of CL#8 is converted into FAR#01 from this Validation & Verification.	
Description of CL raised by VERRA	
Issue The Joint Project Description and Monitoring Report (JPD-MR) defines a value of Bold of 18.54 kg per day (or equivalently 6.77 tonnes per annum) of firewood consumed per stove on average across all seasons. TOOL33, v.02.0 refers to a default value for the average annual consumption of woody biomass of 0.4 tonnes/person/year (wet basis) and the JVR-VR (page 39) states that another project number 3654 reveals a Bold of 6.75 tonnes. However, this project is located in a different country. Action item The VVB must further elaborate how it considered the value used for Bold to be adequate for the project activity considering the referenced default value as per TOOL33, v.02.0 and that the other referenced project is located in Rwanda. Program rule(s) VCS Joint Validation and Verification Report Template v.4.3, Section 3.4.6	
Project participant response	Date: 24/05/2024

- B_{old} is calculated based on direct measurements of daily firewood consumption in kg using digital scales taken as part of the baseline survey.
- The VVB has also verified the fuel consumption measurements during the on-site field visit by conducting interviews with beneficiaries alongside taking direct measurements in the audited households of B_{old}.
- The Project Proponent took 1800 baseline survey points for PAI 1 (including measurements) which is an order of magnitude greater than the methodology requirement.
- Project surveys showed that the communities in the project location are located in a high altitude area (approx. 1000m above sea level) and exposed to low temperatures in the evenings. As a result, the communities not only use their stoves for household cooking, but also for boiling water for the purposes of bathing and drinking. This leads to the increased firewood consumption of project households compared to other regions in India and the default value provided in TOOL33 v.02.0.
- The project conditions here also have some similarity to the climatic conditions of the project region in Rwanda mentioned by the VVB, which resulted in the comparison of B_{old} values.

Documentation provided by project participant

VVB Assessment

Date: 29/05/2024

- a) The value of 18.54 kg/day (equivalent to 6.77 tonnes/annum) was derived from direct measurements of daily firewood consumption using digital scales during the baseline survey. These measurements were verified by the VVB during on-site field visits, which included interviews with beneficiaries and direct measurements of firewood consumption in audited households. During these visits, the VVB team also weighed the firewood at the households using a scale. Additionally, all baseline and end users were interviewed about their firewood use in the baseline scenario. The VVB cross-checked the responses and direct measurements from the audited samples with the baseline database column "AH", "Fuel used per day (Kg)" and concluded that the value of the parameter is acceptable and appropriate.
- b) The project implemented under first PAI is a very specific region like Chintapalle Mandal (Approx 1,100 Meters), Paderu Mandal (Around 900 – 1,200 Meters), GK Veedhi Madal (Around 1,000 Meters), Hukumpeta Mandal (Around 900 – 1,000 Meters), Ananthagiri Mandal (Approx 1,000 meters) etc., within Alluri Sitarama Raju district of Andhra Pradesh, at an altitude of around 900 – 1,200 meters above sea level, where low temperatures are reported. Consequently, stoves are used not only for cooking but also for boiling water for bathing and drinking purposes as well as room heating. This multifaceted usage results in higher firewood consumption compared to other regions in India.
- c) VVB reviewed few other projects which were registered with more or less same Bold values as tabulated below:

Sr. No.	VCS ID	Bold determined by back calculating $B_{y,Saving}$.	Date of Registration	Location
1.	3878	6.244	20/03/2024	India (Rajasthan, Gujarat)
2.	3250	6.117	16/01/2024	India (Assam)
3.	4312	6.752	15/02/2024	Bangladesh
4.	4250	5.222	08/05/2024	India (Meghalaya)
5.	2380	7.635	11/05/2023	Rwanda

To cross check the Temperature variations VVB has checked temperature in the particular region from the Climate change Knowledge Portal, World Bank & Meteologix /36/. Given these conditions VVB concluded that the Bold value observed aligns more closely with the firewood consumption patterns in similar climatic regions, including the project region in Rwanda, despite the geographic differences.

Based on the extensive baseline data, verified measurements, and consideration of local climatic conditions, we find that the Bold value used in the project is adequately justified and aligns with the on-ground realities and specific conditions of the project area.

CL ID	09	Section no.	ER sheet	Date : 13/12/2023
Description of CL				
There are few findings from the technical reviewer as follows: 1. Database of beneficiaries some names are repeating. 2. Some beneficiaries are utilizing gas stoves in the baseline, how are ERs from those being claimed. 3. The majority of the users (50%) used a 'Traditional mud clay double pot' kind of stove however the ICS provided is single pot. How is it ensured that the provided ICS fulfills the requirements of the HHs and the older stoves are not being used. 4. In the distribution database row A46161 onwards there are 7 entries made at the same time and date. The reason for the same is not understood. 5. In cases where the phone number is the same for 2/3 HH how is it ensured that the ICS is able to meet their daily/cooking requirements 6. Row A41936 indicates a family size of 0. Data needs QA/QC. 2576 HHs have a family size of 1 see if the assumption has been used in the ER calculations. 7. How was the baseline study (see page 87 value for B_old) done for winters deemed conservative. The consumption of warm food and stove usage for water heating is also there. Please justify				
Project participant response				Date : 14/12/2023

1. All entries (with photo proof) have unique Aadhaar/ID cards, BPL cards and locations signifying they are unique individuals/families. It should be noted that in communities there are some individuals with the same names. 2. We have checked this point with the field team. No ICS beneficiaries have used gas/LPG in the baseline. The very few entries referred to in the database are incorrect entries, which have been re-checked with the field team and updated in the Distribution database (attached) accordingly. Note that any households that were observed to be using gas/LPG in the baseline were strictly not included in the project. 3. As per the Distribution database, all beneficiaries signed a license agreement stating that their previous stove could be demolished/removed. Further, from the monitoring database (see columns AJ and AN), all users confirmed that they were not using any previous/traditional stoves. From monitoring surveys 100% of users were satisfied with their ICS (see column AC in Monitoring database) and no complaints were made. There was however some feedback from larger families stating that it would be nice to use two ICS. We have taken this feedback into consideration for future distributions. If multiple ICS are given to a single family, this will be accounted for using the parameter n_{HH} (Equation 10 in AMS-II G and Equation 3 in PDMR), which is detailed further on pages 90-91 of the PDMR. 4. The dates/times correspond to the point at which the entries are synced from the local tablet to the cloud database under an internet connection. Multiple entries are synced at a time, usually in the evenings or following morning as the remote areas do not have internet connection. Note that using this date/time in ER calculations is more conservative, as it is later than when the corresponding beneficiary interview and stove distribution is undertaken. 5. In some communities it is not common to have a personal mobile, in which case a group community number is provided. However, all beneficiaries have unique Aadhaar/ID cards, BPL cards and locations signifying they are unique individuals/families. 6. The number 0 has been checked and corrected to 3. Please refer to the corresponding BPL card photos for this family (columns AB and AC) which confirms the family size. 7. During baseline surveys, all respondents were asked to provide the average volume of wood consumed across all seasons, which the project team then measured using weighing scales. Consumption was not specific to a particular day or month.	
Documentation provided by project participant	
VVB Assessment	Date: 14/12/2023

1. The assessment team has confirmed that the end users share common name. However, after reviewing the Aadhar/ID card/BPL Card provided in columns F and G of the distribution database spreadsheet, it was confirmed that there is no double counting or repetitive individuals reported. Hence, the finding is closed.
2. The assessment team confirms that the human error has been rectified in the revised distribution datasheet. No beneficiaries have used gas stoves/LPG in the baseline nor in the monitoring survey which was also confirmed during the site visit. Hence, the finding stands closed.
3. During the audit site visit, the VVB team interviewed end users and confirmed that the project stoves were only given to those end users who agreed to demolish their traditional stoves i.e., mud stove/three stone stove. Additionally, the assessment team visited households and there were no traces of any stove in the kitchen other than the project ICS. It was also cross-checked in the Monitoring database spreadsheet (columns AJ & AN) that no end users were using a traditional stove.
4. During the audit site visit, it was observed that there was no internet connection. Therefore it is understandable that online syncing of the survey data cannot be undertaken immediately and is instead required to be done at a later time along with multiple other entries.
5. The audit team confirms during the site visit that the project populations were not having basic facilities such as road connectivity, network connectivity, education or training, due to which members of the project population either did not have a phone or were unable not use their phone and other essentials properly. Hence, it plausible that a group community contact number is being referred to in the baseline and monitoring databases.
6. The number 0 for the household has been rectified by the Project Proponent in the distribution database. Hence, the finding is closed.
7. VVB confirms that the consumption of wood has been accounted as the average fuel consumption over the year which has been confirmed from the baseline survey sheet provided by the PP. The same has been reflected in the revised PDMR.

Hence, CL#09 is closed.

CL ID	10	Section no.	JPDMR	Date : 22/05/2024
Description of CL raised by VERRA				

Issue

- (a) Section 1.1 does not include the ex-post emission reductions table for the first monitoring period.
- (b) In the audit history table under Section 1.2, the number of years is less than a year. However, the year fraction has not been indicated. The same applies to Section 3.1 of the JVR-VR.
- (c) Under Section 1.7 the entity Cropzone Agro Forestry Private Limited is mentioned as a Project Participant (Role in the Project Section). If so, it shall be included as a second table under Section 1.6. Also, the role of the entity Tramontana Asset Management (page 1 of the JVR-VR) is not mentioned in the submitted documents.
- (d) The ex-ante ER spreadsheet does not include a summary tab for each vintage, average and total for the first crediting period.
- (e) Section 4.1 of the JPD-MR does not include a justification whether one project stove meets the equivalent thermal energy needs of the baseline stove(s), without resorting to the baseline stove.
- (f) Under Section 4.1, a project activity instance (PAI) is not clearly defined, including 'the number of ICS for each instance'. As per the JPD-MR, each distribution per State constitutes one PAI. Consistency is lacking.
- (g) In page 78 of the JPD-MR it is stated that the terminal efficiency will be 20% which is not in line with para. 37 of AMS-II.G.
- (h) In Sections 6.2 and 7.1 the difference between $n_{tot,y,i,j}$ and $n_{0,i,j}$ is not clear. Request you to kindly address the comments

Action item

1. The VVB must ensure that the project proponent addresses points a to h above and updates Sections 1, 4, 6 and 7 of the JPD-MR as needed.
2. The VVB must validate this information and update the JVR-VR as needed.

Program rule(s) or methodology section

VCS Project Description and Monitoring Report Template v.4.3, Sections 1, 4, 6 and 7

Project participant response

Date : 24/05/2024

- a) An ex-post emissions reduction table for the first monitoring period has been added to Section 1.1.
- b) The number of years in Section 1.2 Audit History has been updated as 0.586 to reflect the year fraction for the monitoring period. The same has been updated in the JVR-VR.
- c) Cropzone Agro Forestry Private Limited cannot be included under Section 1.6 as it is not the Project Proponent. It plays a role as the local entity conducting on-the-ground implementation work. Cropzone Agro Forestry Private Limited is a subsidiary of Tramontana Asset Management and this has been added accordingly into the table under section 1.7 of the JPD-MR.
- d) A summary tab for each vintage, average and total has been included in the ex-ante ER spreadsheet provided – see Excel sheet “ICS-01 Ex-Ante vs Achieved Calculations M1 Jun 24” provided.
- e) Based on our monitoring survey sample results (see database) for the first period, it was observed and explicitly confirmed by all beneficiaries that their baseline stove had been removed or demolished, and they were not using any form of baseline stove currently. The VVB confirms both through observations as well as direct discussions with the beneficiaries that none of the households audited have been using any baseline stove. All these findings confirm that the new improved project cookstove met the equivalent thermal energy needs of the baseline stove(s). Language has been added to this effect in Section 4.1.
- f) In Table 1 of the JPD-MR, the number of stoves expected to be distributed per instance is presented. Note that the terminology used here is “regions”, and we do not state anywhere in the document that each state corresponds to an instance. For the first monitoring period 46,160 ICS have been included to form part of Project Activity Instance 1 (and as of August 2023 distribution under PAI 1 had not yet been completed). Some clarificatory text has been added into section 4.1 to improve clarity. Note that when PAI 2 is verified in the next monitoring period it will be in the same state, but a different region given the large size of the state.
- g) AMS-II.G para.37 part (a) from version 13 had been applied - “A default schedule of linear decrease in efficiency up to the terminal efficiency assumed as 20% shall be applied through the life span of the project device” when adopting option 3: water boiling test (WBT). Version 13 of AMS-II.G was the available methodology at the time of submission of the JPD-MR. Since then AMS-II.G version 13.1 has been released (March 2024), which uses a less conservative 25% for the terminal efficiency. As per your request, we have updated the ex-ante calculations in the JPD-MR and calculation sheet to reflect the new terminal efficiency (see Excel sheet “ICS-01 Ex-Ante vs Achieved Calculations M1 Jun 24” provided). Also please note that the terminal efficiency value discussed here is purely used for ex-ante calculation purposes, and is not used in actual emission reductions calculations which result in carbon credit issuance.
- h) Note that the lower case n parameter is a sample based parameter, whereas capitalised N is a population based parameter. This has already been made clear in the JPD-MR section 6.2. Note the table in this section where the parameter $n_{y,i,j}$ is defined, with $n_{tot,i,j}$ and $n_{op,i,j}$ forming part of its definition (see ‘Value applied’). $n_{tot,i,j}$ specifically represents the total number of households sampled, and $n_{op,i,j}$ are those that are operational in the sample. By contrast the capitalised $N_{0,i,j}$ represents the total population of commissioned devices for a

given monitoring period. Hence for the first monitoring period $N_{0,i,j} = 46,160$, whilst $n_{tot,i,j} = 840$ as per the number of samples taken.	
Documentation provided by project participant	
VWB Assessment	Date: 28/05/2024

- (a) VVB confirms that the PP has now included the ex-post emissions reduction table for the first monitoring period in section 1.1 of the JPD-MR version 1.5, which is found to be appropriate.
- (b) VVB confirms that the PP has now revised the number of years in Section 1.2 Audit History as 0.586 to reflect the year fraction for the monitoring period which is found to be appropriate and the same has been updated in the JVR-VR version 2.
- (c) PP has confirmed that Cropzone Agro Forestry Private Limited is not the Project Proponent. It plays a role as the local entity conducting on-ground implementation work which was verified by the VVB during the interviews with the PP and Cropzone Agro Forestry Private Limited during the on-site visit. PP also confirmed that Cropzone Agro Forestry Private Limited is a sister company of the PP, which are both subsidiaries of Tramontana Asset Management and this has been added accordingly into the table under section 1.7 of the JPD-MR version 1.5 which is found to be acceptable.
- (d) PP has revised the ER sheet, by adding a summary tab (“ER Vintage Summary”) in the ER sheet. The information regarding the vintage period, estimated ER VCU, including total and average over the first crediting period, have been added (see tab “ER Vintage Calculation”). Hence, VVB concludes that the revisions made in the revised ER sheet are appropriate & acceptable by the VVB.
- (e) The VVB confirms that the Project Proponent (PP) has demonstrated that the new Improved Cookstoves (ICS) adequately meet the thermal energy needs equivalent to those of the pre-project stoves. The VVB verified this through beneficiary interviews confirming that baseline stoves were either removed or demolished during the installation of the project cookstoves. No baseline stoves were found to exist post-installation. Furthermore, end users confirmed that the project stoves efficiently meet their cooking and other daily requirements, significantly surpassing the performance of their previous traditional baseline stoves. Therefore, the VVB concludes that the new ICS adequately fulfils the thermal energy needs previously provided by the baseline stove. Some further details confirming this point have been added into the section 4.1 of the JPD-MR.
- (f) VVB confirms that Table 1 of the JPD-MR presents the number of stoves expected to be distributed in each instance of the project. The JPD-MR does not indicate anywhere that each state corresponds to an instance, as one state may have multiple instances given its large size. For the current monitoring period 46,160 ICS have been distributed under Project Activity Instance 1 (and as of August 2023, distribution under PAI 1 had not yet been completed) and the same has been mentioned in section 4.1 of the JPD-MR.
- (g) In the JPD-MR, AMS-II.G para.37 part (a) from version 13 had been applied - “A default schedule of linear decrease in efficiency up to the terminal efficiency assumed as 20% shall be applied through the life span of the project device” when adopting option 3: water boiling test (WBT). Note that Version 13 of AMS-II.G was the available methodology at the time of submission of the JPD-MR. Since then AMS-II.G version 13.1 has been released (March 2024), which uses 25% for the terminal efficiency. As per your request, PP has updated the ex-ante calculations in the JPD-MR and calculation sheet to reflect the new terminal efficiency, which corresponds to a reduction in annual efficiency loss of 1.57% (calculated as the initial efficiency minus terminal efficiency of 25% divided by the lifespan of the device). Using this reduction factor, the efficiencies at the start and end of each year have been calculated and averaged to give an average efficiency for each year. The ex-ante ERs have accordingly been updated from 9,497,080 tCO₂e to 9,171,123 tCO₂e (see Excel sheet “ICS-01 Ex-Ante vs Achieved Calculations M1 Jun 24” provided). Therefore, the VVB confirms that the thermal efficiency in section 5.4 (see the page no. 82) of the JPD-MR has been revised and found to be in line with the applied methodology and revised the ex-ante ERs in the JVR-VR.

(h) In the JPD-MR, the notation ' n ' represents the number of Improved Cook Stoves (ICS) based on the sample, with ' n_{tot} ' referring to the total number of ICS sampled, and ' n_{op} ' denoting the number of operational ICS in the sample. The notation ' $N_{o,i,j}$ ' by contrast signifies the total population number of project devices of type i in instance j commissioned over the period. The VVB has reviewed and confirmed that the distinctions made by the Project Proponent (PP) between these parameters are appropriate. Specifically: the lowercase ' n ' is used as a sample-based parameter & the capitalized ' N ' is used as a population-based parameter. Hence, VVB concludes that this distinction aligns with standard statistical and monitoring practices, ensuring clarity and precision in parameter usage throughout the report.

CL ID	11	Section no.	JVVR – 3.1	Date : 22/05/2024
Description of CL raised by VERRA				
Issue Under Section 3.1 of the JVR-VR the project start date has been validated as per the PP's database. Action item 1. The VVB must further substantiate how it validated the project activity by only checking the PP's database and not objective evidence such as records that include the ICS serial number/receipt/user agreement. Program rule(s) or methodology section. VCS Joint Validation and Verification Report Template v.4.3, Section 3.1				
Project participant response				Date : 24/05/2024
- Each entry in the database includes a substantial amount of objective evidence including date/time of submission, ICS serial number, signed user agreement, beneficiary survey information, measurements, photos of each beneficiary with their new ICS, and personal identification records. All of these data points have been automatically synced and recorded onto the cloud database from an Android app which has been purpose built by the Project Proponent for the project. This is an extensive dataset validating the project and the VVB has described in their JVR-VR of not having seen this level of detail and information provided on each beneficiary in other projects. - The date/time of the first entry has been automatically synced to the cloud from the local tablet, and it is evident from the dataset that this specific date/time is 21st January 2023. - Also attached is the meta data for the photos from the first database entry. As can be seen from the meta data, the date recorded of 21st Jan 2023 corresponds to the project start date.				

Documentation provided by project participant	
VVB Assessment	Date: 28/05/2024
The VVB verified the distribution database (including project start date) provided by the PP, which contains a substantial amount of objective evidence including date and time of submission, ICS serial numbers, signed user agreements, National ID proof of end user, Photographs of End User, Below poverty line card issued by Government of Andhra Pradesh under National food security Act, 2013, Photographs with their new ICS, and personal identification records. Additionally, during the on-site visit, the VVB checked the Android application developed by the PP, which is designed to record and automatically sync all data points including date/time of submission. It was observed that this application seamlessly automatically synced data to the cloud from each local tablet, minimising any possibility of manual alterations or errors. To further verify the project start date, the VVB assessed the metadata attached to the photos (beneficiary with ICS, national ID card front and back, BPL card front and back, community certificate front and back) from the first recorded entry which was synced from the tablets. The VVB is providing this metadata to Verra in an Excel spreadsheet. The data confirms that the date and time of creation of these photos was the 21st January 2023 (column E in the metadata spreadsheet). Consequently, based on the distribution database records and metadata, the VVB confirms that the project activity's start date is 21st January 2023, and this has also been reflected in the Joint Validation and Verification Report (JVVR) version 2.).	

CL ID	12	Section no.	JPDMR – 1.4.1	Date : 22/05/2024
Description of CL raised by VERRA				
Issue Under Section 1.4 of the JPD-MR it is not clear how it is ensured that each added project activity instance of the same project activity may not be spread across more than one project if they are within ten kilometers of each other), as required by Section 3.6.8 of the VCS Standard v.4.7. Action item 1. The VVB must ensure that the project proponent includes in Section 1.4 of the JP-MR detailed information on how each added PAIs complies with Section 3.6.8 of the VCS Standard. 2. The VVB must validate this information and update the JVR-VR as needed. Program rule(s) or methodology section VCS Standard v.4.7, Section 3.6.8				

Project participant response	Date: 24/05/2024
- Additional text has been included in section 1.4 of the JPD-MR to provide more clarity on meeting such condition. - The project proponent has ensured that all project activity instances to date (and in the future) are recorded under this one single project. Note that there are currently no additional cookstove projects in India listed in the Verra registry by Tramontana.	
Documentation provided by project participant	
VVB Assessment	Date: 28/05/2024
VVB confirms that the PP has now included wording in section 1.4 of the JPD-MR that each added project activity instance of the same project activity may not be spread across more than one project if they are within ten kilometers of each other as per the section 3.6.8 of the VCS Standard v.4.7 and also confirms that the project activity instance as of the current monitoring period is under a single project. The VVB has validated the requirement based on the geolocations of beneficiaries recorded in the project database, and confirm that they are all within a single project and have communicated validation of this condition accordingly in the table in section 3.1 of the JVR-VR.	

CL ID	13	Section no.	JPDMR 3.1	Date : 22/05/2024
Description of CL raised by VERRA				
Issue Section 3.1 of the JPD-MR does not include applied CDM Methodological TOOL30 and the applied CDM Methodological TOOL33. Action item 1. The VVB must ensure that the project proponent includes all the applicable TOOLS and applicability criteria under Sections 3.1 and 3.2 of the JPD-MR. Program Rule(s) VCS Project Description and Monitoring Report Template v.4.3, Sections 3.1 and 3.2				
Project participant response				Date : 24/05/2024
Section 3.1 and 3.2 of the JPD-MR have been updated to include these tools and how their criteria are satisfied.				

Documentation provided by project participant	
VVB Assessment	Date: 28/05/2024
VVB confirms that the PP has now included all-applied tools in section 3.1 of the JPD-MR and their applicability conditions, as well as how the criteria are fulfilled in section 3.2 of the JPD-MR version 1.5. Therefore, VVB concluded that the revised section 3.1 of the JPD-MR is in line with the JPD-MR template, and hence, is acceptable to the VVB. VVB has added the assessment on Tool 30 by verifying this tool was used to calculate the project specified value for fNRB under option (b) of the applicability conditions. For the PA, the fNRB is calculated as 0.894 in line with equation 1, 2, 3& 4 of the tool. Assessment of the fNRB is provided under section 3.4.6 of this report. The tool is used by the Project Proponent to calculate project specific fNRB values which has been checked in section 5.4 of the JPD-MR/01/. For future PAs as well, Tool 30 will be applicable. Hence, the criterion is found to be met. Similarly, for Tool 33 VVB confirmed that the PP has determined the project start date is prior to 10th March 2025 and therefore, the default value of 0.15 used from CDM TOOL33 for the efficiency of pre-project cooking devices is valid. Any project activity occurring after 10th March 2025, will use the latest default value available. VVB has revised the section 3.4.2 “Applicability” of JVVR version 2, dated 14 June 2024.	

CL ID	14	Section no.	JPDMR 3.2	Date : 22/05/2024
Description of CL raised by VERRA				
Issue In order to confirm the applicability criterion under para. 5 of AMS-II.G, v.13.1, Section 3.2 of the JPD-MR refers to a reports from 1970-1997. However, it is not clear how it has been demonstrated that non-renewable biomass has been in use in the region from 1997 onwards. Action Required 1. The VVB must ensure that the project proponent describes in detail how they demonstrate that Non-renewable biomass has been used in the project region since 31 December 1989 (until present). 2. The VVB must validate this information and update the JVR-VR as needed. Program Rule(s) AMS-II.G, v.13.1, para. 5				
Project participant response				Date : 24/05/2024

- The point made in response to this condition was to demonstrate that non-renewable biomass has been used since at least before 1989, and therefore the justification in section 3.2 had focused on demonstrating this.
- As per the request of the technical reviewer, for the period beyond 1997 data sources confirm that use of non-renewable biomass also increased. The Global Forest Watch report between 2001 to 2023 states that India lost 2.33Mha of tree cover, equivalent to a 6% decrease in tree cover since 2000 (reference: [https://www.globalforestwatch.org/dashboards/country/IND/?category=forest-change&location=WYJb3VudHJ5IiwSU5EIIO%3D\).India](https://www.globalforestwatch.org/dashboards/country/IND/?category=forest-change&location=WYJb3VudHJ5IiwSU5EIIO%3D).India) Deforestation Rates & Statistics | GFW ([globalforestwatch.org](https://www.globalforestwatch.org)). This point has been included in section 3.2 of the JPD-MR.

Documentation provided by project participant

VVB Assessment

Date: 28/05/2024

The Project Proponent has documented the use of non-renewable biomass in the project area since at least before 1989, with data sources confirming this usage until 1997. In the revised JPD-MR version 1.5, the PP explicitly states the continued use of non-renewable biomass up to 2023, supported by referenced data (see section 3.2 of JPD-MR). This data was checked by the VVB and found to be acceptable. Therefore, VVB confirms that the PP has adequately demonstrated the continuous use of non-renewable biomass in the project region from December 31, 1989, until the present. This information has also been incorporated into the updated JVR-VR version 2 by the VVB.

CL ID	15	Section no.	JPDMR 3.4.5	Date : 22/05/2024
Description of CL raised by VERRA				
Issue Section 3.4.5 of the JVR-VR does not explain how the VVB verified that the project ICSs are offered free of cost to end users.				
Action item The VVB must elaborate how it verified that the cookstoves are implemented and distributed free of cost to the end users.				
Program Rule(s) VCS Joint Validation and Verification Report Template v.4.3, Section 3.3.5				
Project participant response				Date : 24/05/2024

- Under clause 1a of the user agreement during distribution (see links in database), it states “we will not charge to you for the License to use the ICS”. This binding agreement is signed by both each beneficiary and the representative of the Cropzone field team conducting the survey.
- Through interaction with project beneficiaries, during the audit, the VVB asked multiple beneficiaries to confirm that their project ICS had been provided free of charge. The beneficiary directly confirmed this in the local language.

Documentation provided by project participant

VVB Assessment

Date: 29/05/2024

The VVB has mentioned in section 3.4.5 of the JVR-VR that they confirmed the project Improved Cookstoves (ICSs) were distributed free of cost to the beneficiaries. This was confirmed upon directly by interviewing beneficiaries during the audit visit to the project site. Additionally, the VVB has checked the signed user agreements for the ICS, where it clearly says in clause “1a” that the beneficiaries won't be charged for using the ICS. This agreement was signed by each beneficiary and the Cropzone field team representative who conducted the survey. Hence, VVB concluded that the stoves were given to the beneficiaries without any cost.

CL ID

16

Section no.

Date : 22/05/2024

Description of CL raised by VERRA

Issue

For the calculation of fNRB, there is lack of information on:

- (a) If it has been estimated using the most recent historical year for which data is available.
- (b) How H has been calculated as per Section 3.1 of the TOOL30, v.04.0;
- (c) How RB has been calculated as per Section 3.2 of the TOOL30, v.04.0;
- (d) Further, there is no information on:
 - i. the comparison of the calculated value of fNRB against the values for fNRB reported in relevant scientific literature (including the minimum requirements as per para. 6-a-i of the TOOL30) and justify any differences and;
 - ii. the cross-checks performed as required by para.13 of the TOOL30, v.04.0. Further, they state in page 59 that they use version 03.0. and spreadsheet not provided, no info of each of the values used.
 - iii. The calculation spreadsheet has not been provided.

Action Required

1. The VVB must ensure that the project proponent further elaborates how the fNRB has been calculated in line with the requirements of TOOL30, v.04.0 and addresses points a to c above.
2. The VVB must verify this information and update the JVR-VR as needed.
3. The calculation spreadsheet shall be provided for further reference. The fNRB steps shall also be included in Section 5.4 of the VCS-PD

Program Rule(s)

TOOL30, v.04.0.

Background information

TOOL30, v.04.0, para. 6 (b): The project participants shall compare and analyse the calculated values against the values for fNRB reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least: (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.

Project participant response

Date : 24/05/2024

- The f_{NRB} calculation sheet has already been supplied – see Excel sheet titled “ICS-01 f_{NRB} Calculation M1 Oct 23”
- Additional text has also been added in section 5.4 of the JPD-MR outlining the f_{NRB} calculation

Further, in line with CDM TOOL30, the calculated f_{NRB} has been validated by comparing it to f_{NRB} values from projects in similar regions and with the literature listed in paragraph 6(b) of TOOL30: *Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.*

The table below summarises recent registered cookstove projects in India under Verra VCS, and their calculated f_{NRB} :

Project ID	Registration Date	f_{NRB} Value	Project Location
VCS 3240	15/08/2023	0.8714	Odisha
VCS 3413	04/01/2024	0.913	Maharashtra
VCS 3769	18/12/2023	0.8755	Madhya Pradesh
VCS 2664	23/11/2022	0.852	Assam
VCS 3161	10/01/2023	0.95	Telangana

The climate of Assam, Odisha and Telangana have similar characteristics to Andhra Pradesh which is our project location for Project Activity Instance 1. Our f_{NRB} is comparable to other projects in these regions which have all been registered under Verra, as observed from the table.

Please note that for this project a new f_{NRB} value will be calculated for each new state in which a project activity instance may occur.

The literature referenced in CDM TOOL30, Bailis et al. (2015), reports an f_{NRB} value of 23-24% for India. This lower value as compared with our calculated value (and those in the table above), can be attributed to a number of reasons:

- As stated in CDM TOOL30 v.4.0: “in the case of ex ante calculation of f_{NRB} , the parameter f_{NRB} shall be estimated using the most recent historical year for which data is available”. In the case of Bailis et al. (2015) paper, the FAO documentation used to calculate forest cover is dated 2010, which is older than the more recent 2019 data used for our calculations.
- Although Bailis et al. (2015) used subnational estimates to determine the fuelwood consumption in urban and rural parts of India, the f_{NRB} value provided in the main text of the paper is a country-wide average. This encompasses all states of India, many of which have lower f_{NRB} values owing to their low forest cover and firewood usage. The f_{NRB} value for our first project activity instance was calculated using data specific to Andhra Pradesh, and therefore is a more accurate representation for the project area than a country wide estimate.

Documentation provided by project participant

VVB Assessment

Date: 28/05/2024

VVB confirms that the PP has now included in the JPD-MR and 'ICS-01 fNRB calculation sheet' which has been provided that the

- Calculation of H as per Section 3.1 of the TOOL30, v.04.0 and
- Calculation of RB as per Section 3.2 of the TOOL30, v.04.0, which are found to be appropriate.

PP also provided the comparison of the calculated value of fNRB against the values for fNRB reported in relevant scientific literature (including the minimum requirements as per para. 6-a-i of the TOOL30). The literature referenced here in CDM TOOL30, Bailis et al. (2015), reports an fNRB value of 23-24% for India. This is lower than the calculated value for the project for a number of reasons:

- As per CDM in CDM TOOL30 v.4.0: "in the case of ex ante calculation of fNRB, the parameter fNRB shall be estimated using the most recent historical year for which data is available". In the case of Bailis et al. (2015) paper, the FAO documentation used to calculate forest cover is dated 2010, which is older than the more recent 2019 data used for the current calculations made by the PP.
- In line with para 6 of the tool 30 version 4 , which required comparing the fNRB value against published literature, specifically the Bailis et al. (2015), the fNRB value provided in the main text of the paper is a country-wide average. This encompasses all states of India, many of which have lower fNRB values owing to their low forest cover and firewood usage. The fNRB value for the first project activity instance was calculated by the PP using data specific to Andhra Pradesh, and therefore is a more accurate representation for the project area than a country wide estimate.

Additionally, VVB has checked few of the projects registered during 2022 to 2024 under Verra in different states of India which have similar climatic conditions as Andhra Pradesh, where the current project activity is located. The details of the registered projects with their registration date, fNRB values and project location are provided in the table below.

Project ID	Registration Date	fNRB Value	Project Location
VCS 3240	15/08/2023	0.8714	Odisha
VCS 3413	04/01/2024	0.913	Maharashtra
VCS 3769	18/12/2023	0.8755	Madhya Pradesh
VCS 2664	23/11/2022	0.852	Assam
VCS 3161	10/01/2023	0.95	Telangana

Therefore, from the table it is clear that the fNRB calculated in the project activity is comparable with these projects registered in India.

VVB has checked the calculation spreadsheet (which has been provided) for the fNRB steps and found it to be appropriate, and the same has been elaborated in section 5.4 of the JPD-MR.

The JVR-VR has also been updated accordingly in the table in section 3.4.6.

The calculation sheet ("ICS-01 fNRB Calculation M1 Oct 23") has been provided.

CL ID	17	Section no.	JVVR 3.4.6	Date : 22/05/2024
Description of CL raised by VERRA				
Issue				
Section 3.4.6 of the JVR-VR lacks information on how the applied 15% efficiency for the baseline stoves is suitable to all the baseline stoves replaced under the project activity.				
Action item				
1. The VVB must sufficiently elaborate how the value of 15% is applicable to all the baseline stoves replaced under the project activity.				
Program rule(s)				
VCS Joint Validation and Verification Report Template v.4.3, Section 3.46				
Project participant response				Date : 24/05/2024
- The efficiency of a pre-project cooking device is defined in CDM TOOL33 5.6 17(a): “For a three-stone fire using firewood (not charcoal), or a cookstove with no improved combustion air supply or flue gas ventilation (i.e. without a grate or a chimney), the default value is 0.15.” As recorded in the distribution database, all beneficiaries either had a mud clay single/double pot stove or three-stone fixed stove using firewood. Hence, a value of 0.15 is applied. - It can also be noted that the 0.15 value for such pre-project stove devices is conservative, for instance the research sources in the table below present a lower value:				
Source		Stove-Type	Efficiency	
IJISSET V3 I12 05.pdf		Traditional mud clay	8.02%	
Emission factors of wood and charcoal-fired cookstoves - ScienceDirect		Three-stone and mud clay stoves	5-15%	
Documentation provided by project participant				

VVB Assessment	Date: 28/05/2024
VVB has now updated the JVR-VR and elaborated that the applied 15% efficiency for the baseline stoves is suitable to all the baseline stoves replaced under the project activity. The VVB has checked the database of surveys conducted by the PP for the baseline scenario which shows that all beneficiaries either had a mud clay single/double pot stove or three-stone fixed stove using firewood. According to CDM TOOL33 5.6 17(a): “For a three-stone fire using firewood (not charcoal), or a cookstove with no improved combustion air supply or flue gas ventilation (i.e. without a grate or a chimney), the default value is 0.15.” Along with this, the VVB also checked the sources such as IJSET V3 I12 05.pdf and Emission factors of wood and charcoal-fired cookstoves - ScienceDirect, where the maximum efficiency is found to be 15% for such baseline stoves. Therefore, VVB confirms that the applied 15% efficiency for the baseline stoves is found to be conservative and hence accepted.	

CL ID	18	Section no.	JPD MR 1.11 & 5.4	Date : 22/05/2024
Description of CL raised by VERRA				
Issue The tables under Sections 1.11 and 5.4 of the JPD-MR and Section 5.2 of the JVR-VR do not report the vintages in ranges (E.g., day-month-year to day-month-year). Action item 1. The VVB must ensure that the project proponent reports the vintages in ranges under Section 5.4 of the JPD-MR. 2. The VVB must report the vintages in ranges under the table in Section 5.2 of the JVR-VR. Program rule(s) VCS Joint Project Description and Monitoring Report Template v.4.3, Section 5.4 VCS Joint Validation and Verification Report Template v.4.3, Section 5.2				
Project participant response				Date : 24/05/2024
Section 1.11 and 5.4 of the JPD-MR have been updated to include the vintages in ranges.				
Documentation provided by project participant				
VVB Assessment				Date: 28/05/2024

Section 1.11 and 5.4 have been revised accordingly to include vintages and VVB is satisfied with the calculations.

All calculations underpinning these tables are provided in the ER sheet ("ICS-01 Ex-Ante vs Achieved Calculations M1 Jun 24").

Table 2.CAR from this validation

CAR ID	01	Section no.	3.2	Date : 06/10/2023
Description of CAR				
In section 3.2 of the PDMR, PP is requested to include all the applicable conditions as per the applied methodologies by mentioning whether they are applicable or not for the project activity				
Project participant response				Date : 25/10/2023
We have added the two remaining applicability conditions into the PDMR in section 3.2.				
Documentation provided by project participant				
VVB Assessment				Date: 06/12/2023
PP has added all applicability conditions as per the methodologies applied in the project activity. Hence, the finding is Closed.				

Table 3.FAR from this validation

FAR ID	01	Section No.	-	Date : 08/12/2023
Description of FAR				
Ex-ante Emission Reductions provided on the Verra page and in the PDMR are inconsistent. The estimated ERs are to be made consistent between the project documents and project activity web page. VVB to verify the same in the subsequent verification				
Project participant response				Date : DD/MM/YYYY
Documentation provided by project participant				
VVB Assessment				Date: DD/MM/YYYY

FAR ID	02	Section No.	-	Date : 08/12/2023
Description of FAR				
For the current project activity instance, the Alluri Seetharama Raju district has been covered in the baseline survey. A baseline survey has to be conducted before the implementation of any new PAIs in compliance with the applied methodology section 9.1.				
Project participant response				Date : DD/MM/YYYY
Documentation provided by project participant				
VVB Assessment				Date: DD/MM/YYYY

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